Genetec Mission Control™ Deployment Guide 3.4.1.0

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Document last updated: August 04, 2026

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Document information

Document title: Genetec Mission Control™ Deployment Guide 3.4.1.0

Original document number: EN.705.007-V3.4.1.0(1)

Document number: EN.705.007-V3.4.1.0(1)

Document update date: August 04, 2026

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About this guide

This guide describes the various deployment models available for the Genetec Mission Control™ Collaborative Decision Management System, and provides instructions on how to install and deploy Mission Control System.

This guide supplements the *Security Center Administrator Guide* and the *Security Center User Guide*. For more information, see the TechDoc Hub.

Notes and notices

The following notes and notices might appear in this guide:

- **Tip:** Suggests how to apply the information in a topic or step.
- **Note:** Explains a special case or expands on an important point.
- **Important:** Points out critical information concerning a topic or step.
- **Caution:** Indicates that an action or step can cause loss of data, security problems, or performance issues.
- **Warning:** Indicates that an action or step can result in physical harm, or cause damage to hardware.

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Part I

Installation and deployment options

This part includes the following chapters:

- Chapter 1, "Mission Control overview" on page 2
- Chapter 2, "Specifications for Mission Control deployment" on page 6
- Chapter 3, "Deployment considerations" on page 15
- Chapter 4, "Installing Mission Control" on page 53
- Chapter 5, "Staging to production environment" on page 81
- Chapter 6, "Upgrading Mission Control and RabbitMQ" on page 85

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Mission Control overview

This section includes the following topics:

- "What is Mission Control?" on page 3
- "What is a Mission Control incident?" on page 5

What is Mission Control?

Genetec Mission Control™ is a collaborative decision management system that provides organizations with new levels of situational intelligence, visualization, and complete incident management capabilities. It allows security personnel to make the right decision when faced with routine tasks or unanticipated situations by ensuring a timely flow of information. To learn more about Genetec Mission Control™, refer to the Genetec™ resource center.

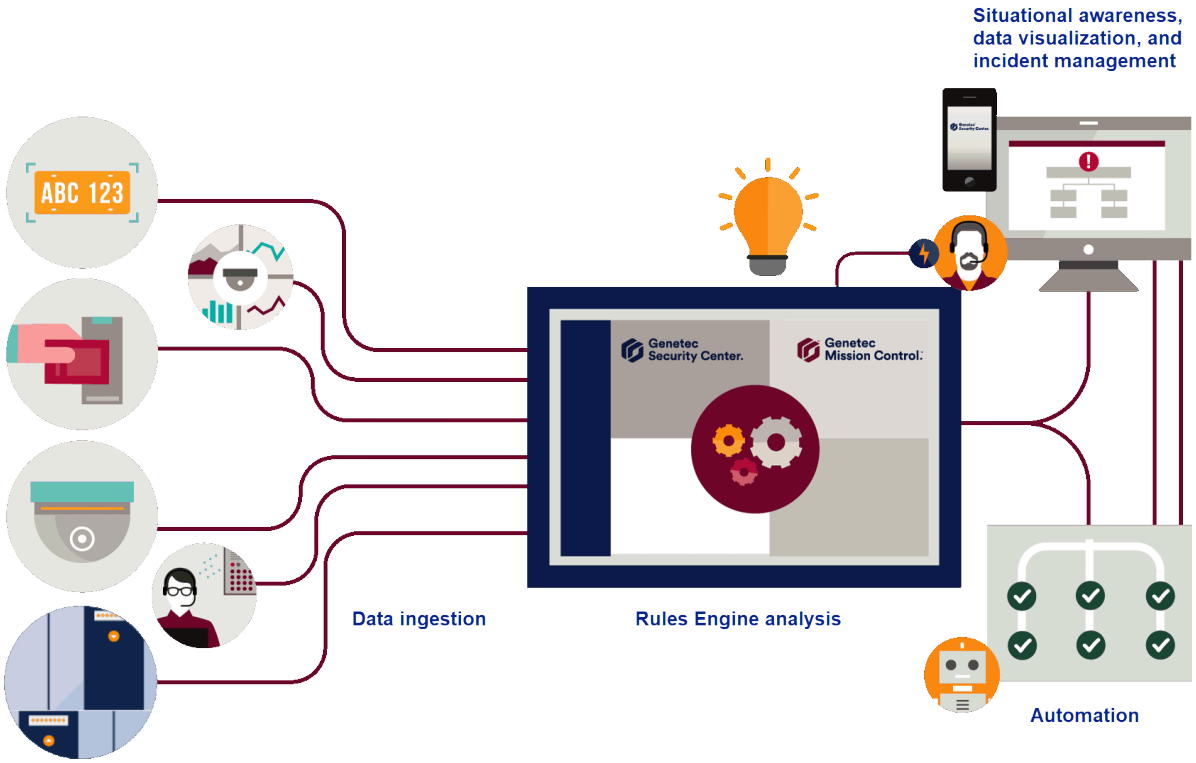
Genetec Mission Control™ user experience is centered around the operator, providing clarity and focus. Events with the highest priority are processed based on response policies and standard operating procedures, eliminating uncertainty.

Situational awareness is created through dynamic map views, contextual details, and real-time status updates. The Mission Control interface provides all relevant data for planning and incident response on a single screen, or across multiple displays, based on the operator's preference.

With thousands of events competing for attention every day, manually reviewing each situation is impossible. The Rules Engine of Mission Control helps you separate real threats from false alarms. It qualifies events and classifies them into incidents, allowing your team to focus on situations requiring their attention.

Mission Control can also automate responses to routine situations, such as equipment maintenance or employee screenings, without intervention from security operators. Its intuitive drag-and-drop interface means you can easily create and expand automation workflows to adapt to new situations, without custom development.

Here is the incident life cycle managed by Mission Control:



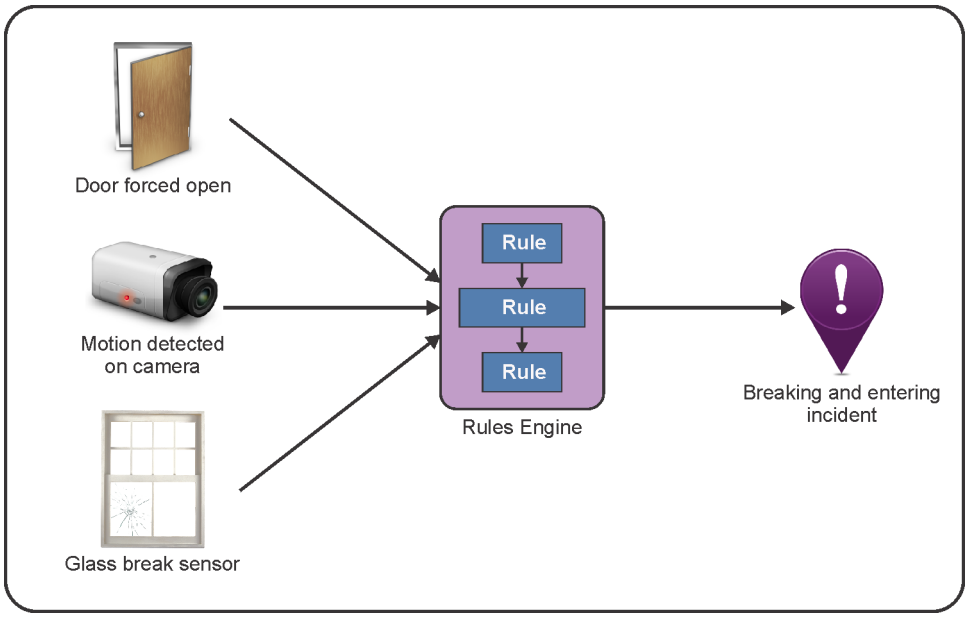
Watch this video to learn more. Click the **Captions** icon (**CC**) to turn on video captions in one of the available languages.

NOTE: Although we do our best to keep our videos current, the information presented in this video might become outdated with each new release. If you find anything wrong with this video, feel free to contact us.

What is a Mission Control incident?

A Genetec Mission Control™ incident is an undesirable or unusual situation that needs investigation and resolution, or a routine, scheduled task that requires monitoring.

An incident can consist of one or more alarms and events, as shown in the following example:



In this example, when three separate events occur at the same time (*Door forced open*, *Motion detected* on camera, and *Glass break*), the *Breaking and entering* incident is automatically triggered. Incidents can be triggered by multiple possible sources, and they are usually not all active at the same time. The rules for automatic incident triggering are configured by an administrator.

If witnesses report problems that do not automatically trigger incidents, security operators can trigger incidents manually.

Incidents go through a series of preconfigured states, which depend on operator actions and other events that occur in the system:

1. When an incident is created, it is in the *New* state.
2. When an operator responds to the incident, the state changes to *In Progress*.
3. After all of the procedure steps are completed, or when the system events and alarms have all been acknowledged, the state changes to *Resolved*.
4. The incident can then be changed to *Closed*.

NOTE: Genetec Mission Control™ incidents are different from Security Center incidents. The Security Center incident can only record video. The Config Tool settings found in **System** > **General settings** > **Incident categories** only apply to Security Center incidents (video only), not to Mission Control incidents.

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Specifications for Mission Control deployment

This section includes the following topics:

- "Mission Control 3.4.1.0 system requirements" on page 7
- "Factors that affect Mission Control performance" on page 8
- "Ports used by applications in Mission Control" on page 10
- "Product compatibility for Mission Control 3.4.1.0" on page 13
- "Supported languages in Mission Control 3.4.1.0" on page 14

Mission Control 3.4.1.0 system requirements

Systems running Genetec Mission Control™ 3.4.1.0 must meet or exceed the *recommended* Security Center server requirements.

For Security Center server metrics, refer the Security Center server requirements topic in the *Security Center System Requirements Guide*.

For the latest up-to-date compatibility information and a list of other compatible products, see Product compatibility for Mission Control.

Factors that affect Mission Control performance

The number of incidents Genetec Mission Control™ can handle per minute depends on the number of incident types, complexity of the incident detection rules, automation workflows, number of active incidents, number of active users and the rate at which events are generated in the system.

The performance metrics of your Mission Control are impacted by the following:

- **Number of events generated per second:** An event can be anything as simple as an access-granted event or a motion detection event.
- **Number of incident types configured:** An incident type entity represents a situation that requires specific actions to resolve it. The incident type entity can also be used to automate the incident detection in Genetec Mission Control™ and to enforce the standard operating procedures that your security team must follow.

  The configuration of your incident types impact performance and also includes the following:

  - Configuration of incident triggers or rules for triggering an incident.
  - Complexity of the automation workflows.

    For example, a **Wait for event** activity in the automation workflow would impact performance.
- **Number of incidents triggered per minute:** This refers to the number of active incidents triggered per minute.
- **Number of incidents active in your system:** The total number of active incidents in your system at any given moment.
- **Number of concurrent user connections:** The number of simultaneous user connections handled by the server. The server can handle only 300 concurrent user connections.
- **Other factors:**

  - Performance of RabbitMQ
  - Performance of your SQL Server
  - Number of servers
  - Number of clients: Includes workstations, Genetec Mission Control Web API clients, and mobile apps.

Disk usage

- The average size of an incident type configuration is approximately 5MB. If you have 100 incident types, you need 500MB of database space.
- The average size of an active incident on disk is approximately 50 KB. If you have 3,000 incidents per day, you need 13.5 GB of disk space for 90 days of retention.
- For RabbitMQ to function optimally, a minimum of 50MB of disk space must be free.

Configuring your system architecture

Based on your requirements, you can build your system architecture using Security Center recommended server or high-end server specifications.

For example, if you have 1 - 4 operators and expect a minimum level of automation in your incident workflows, you can build your system architecture using the Security Center recommended server specifications.

If you have 5 or more operators, expect a large level of automation in your incident workflows with multiple servers. You should build your system architecture using the Security Center high-end server specifications.

TIP: For information about Security Center server metrics, see Security Center server requirements.

Your system performance depends on your setup. For more details on deployment architectures and their requirements, see Deployment options overview on page 19.

Ports used by applications in Mission Control

For Genetec Mission Control™ to work properly, you must create firewall rules to allow proper communication between the various services.

The following table lists the default network ports used by core applications in Mission Control. You can also view the network diagram.

Application	Inbound	Outbound	Port usage
Client applications (Security Desk, Config Tool)		TCP 5500	Directory communication
		TCP 8012	Map background web requests to Map Manager
		TCP 443	Genetec Portal communication for Genetec Advantage validation and feedback Authentication role communication (HTTPS) Web API SDK (HTTPS)
		TCP 5671	RabbitMQ SSL Port (Security Desk)
RabbitMQ	TCP 5671		RabbitMQ SSL Port
	TCP 4369		Erlang Port
	TCP 15671		HTTPS API port
	TCP 25672		Node Clustering Port
Incident Manager	TCP 5500	TCP 5500	Genetec Server/Directory communication
	TCP 4502	TCP 4502	Replaces TCP 5500 for roles originally created in Security Center 5.3 and earlier
	TCP 80	TCP 80	REST/Server Admin communication
	TCP 443	TCP 443	Secured REST/Server Admin/Web Server Authentication role communication (HTTPS) Web API SDK (HTTPS)
		TCP 1433	Outgoing connections to the SQL Database Engine hosted on another server. **NOTE:** Only required for roles that must connect to a database on another server. Not required if SQL Server is running on the same machine.
		UDP 1434	Outgoing connections to the SQL Server Browser service for SQL Server connection information. **NOTE:** Only required for roles that must connect to a named database instance on another server. Not required for roles configured to connect to their database using a specific port.

Application	Inbound	Outbound	Port usage
		TCP 5671	RabbitMQ SSL Ports
		TCP 15671	RabbitMQ HTTPS API port
Directory	TCP 4502	TCP 4502	Genetec™ Server communication
	TCP 80	TCP 80	Directory Web Server communication
	TCP 443	TCP 443	Directory secured web server communication
	TCP 5500		Client Connection
		TCP 5671	RabbitMQ SSL Port
		TCP 9550	Web API SDK
Web API SDK	TCP 9550		API secured web server communication **NOTE:** If this port is blocked, the Directory forwards the requests to the expansion server hosting the Incident Manager role using port 443.
		TCP 443	Directory Secured Web Server communication
		TCP 5671	RabbitMQ SSL Port
Mobile Server	TCP 5500	TCP 5500	Genetec Server/Directory communication
	TCP 4502	TCP 4502	Replaces TCP 5500 for roles originally created in Security Center 5.3 and earlier
	TCP 80	TCP 80	REST/Server Admin communication
	TCP 443	TCP 443	Secured REST/Server Admin/Web Server Authentication role communication (HTTPS) Web API SDK (HTTPS)
		TCP 1433	Outgoing connections to the SQL Database Engine if hosted on another server
		UDP 1434	Outgoing connections to the SQL Server Browser service if on another server
Incident Document Service	TCP 5500	TCP 5500	Genetec Server/Directory communication
	TCP 4502	TCP 4502	Replaces TCP 5500 for roles originally created in Security Center 5.3 and earlier
	TCP 80	TCP 80	REST/Server Admin communication
	TCP 443	TCP 443	Secured REST/Server Admin/Web Server Authentication role communication (HTTPS) Web API SDK (HTTPS)

Application	Inbound	Outbound	Port usage
		TCP 1433	Outgoing connections to the SQL Database Engine if hosted on another server
		UDP 1434	Outgoing connections to the SQL Server Browser service if on another server
Report Manager		TCP 5671	RabbitMQ SSL Port
Web App Server	TCP 5500	TCP 5500	Genetec Server or Directory communication
	TCP 4502	TCP 4502	Replaces TCP 5500 for roles originally created in Security Center 5.3 and earlier
	TCP 80	TCP 80	REST/Server Admin communication
	TCP 443	TCP 443	Secured REST/Server Admin/Web Server Authentication role communication (HTTPS) Web API SDK (HTTPS)
		TCP 1433	Outgoing connections to the SQL Database Engine if hosted on another server
		UDP 1434	Outgoing connections to the SQL Server Browser service if on another server
External SQL Server	TCP 1433		Incoming connections to the SQL Database Engine from roles on other servers
	UDP 1434		Incoming connections to the SQL Server Browser service for SQL Server connection information

Product compatibility for Mission Control 3.4.1.0

Genetec Mission Control™ 3.4.1.0 is compatible with specific versions of Security Center and other products.

Mission Control 3.4.1.0 is compatible with Security Center Enterprise, versions 5.14.0.0 and 5.14.1.0.

For the latest up-to-date compatibility information and a list of other compatible products, see Product compatibility for Mission Control.

Supported languages in Mission Control 3.4.1.0

Genetec Mission Control™ 3.4.1.0 software is available in the following languages.

- Arabic
- Dutch
- English
- Finnish
- French
- German
- Italian
- Japanese
- Korean
- Spanish
- Thai
- Turkish

Genetec Mission Control™ Web API is available in English.

3

Deployment considerations

This section includes the following topics:

- "Evaluation checklist for Mission Control deployment" on page 16
- "Components involved in Mission Control deployment" on page 17
- "Deployment options overview" on page 19
- "Basic deployment option" on page 20
- "Security Center Directory deployment options overview" on page 22
- "Incident Manager deployment options overview" on page 27
- "SQL Server deployment options overview" on page 32
- "RabbitMQ deployment options overview" on page 34
- "Configuring high availability for RabbitMQ" on page 39
- "Assigning SQL permissions on remote SQL Servers" on page 43
- "Determining the best deployment architecture for you" on page 52

Evaluation checklist for Mission Control deployment

To manage Genetec Mission Control™ deployment effectively using Mission Control, it is recommended that you understand your existing environment and evaluate the requirements for tolerance to failures.

Evaluation of current assets and requirements analysis

To evaluate the scope of fault tolerance required and what can be accomplished using your existing systems, you must ask the following questions:

- ☐ What are the specifications of machines or virtual machines currently available?
  - ☐ How many machines or virtual machines are at hand?
  - ☐ How powerful are those machines in terms of computation (CPU)?
  - ☐ How much random access memory (RAM) do they have?
  - ☐ What kind of disks are those machines equipped with: solid-state drives (SSDs) or hard disks?
  - ☐ How much disk space do they have?
  - ☐ Are the machines located in the same rack?
  - ☐ What type of maintenance gets done on those machines?
  - ☐ What exposure do they have to external communications?
- ☐ What does the network environment look like for the deployment?
  - ☐ Are there certain points of congestion in the network?
  - ☐ Is it a geographically distributed network?
  - ☐ Are there any higher latency points in the network or between networks?
  - ☐ Is frequent maintenance required between certain points of your network?
  - ☐ What does the redundancy look like?
  - ☐ Are there any firewalls, Intrusion Detection Systems (IDS) or Intrusion Prevention Systems (IPS) managing part of the communications?
- ☐ What is your most important concern between availability and consistency?
  - **Consistency:** You prefer losing availability of a service over losing data.
  - **Availability:** You prefer losing data to preserve availability or to restore it faster.

  In Mission Control, if downtime happened, would you rather restore service faster (availability) or lose incidents triggered in the past (consistency)?

Components involved in Mission Control deployment

For effective deployment, you must understand the roles of the multiple components of Genetec Mission Control™.

Incident management through Mission Control involves the following components:

- **Incident Manager role:**
  - Installed as part of Mission Control.
  - It runs all the incident business logic, managing incidents within the system.
  - Can be deployed on any Security Center server or expansion server.
  - Supports *failover* mode.
  - Incident Manager role needs to be created to use the component.
- **Security Center Directory:**
  - Main supervisor service of all Security Center roles.
  - Performs the following functions:
    - Authenticates users.
    - Manages entities.
    - Maintains health monitoring stats of the various roles.
    - Pushes events from all Security Center components to RabbitMQ for Mission Control.
  - Can be deployed on any Security Center server or expansion server.
  - Supports failover mode.
- **RabbitMQ:**
  - Third-party message broker allowing communications between the various components of Mission Control.
  - Acts as a buffer to prevent data loss if the Incident Manager role ever goes down.
  - Serves as a load-leveling mechanism preventing high burst of events and data coming from the Security Center system that could cripple the Incident Manager role.
  - Enables a replay of past events.
  - It's the only component that supports *high availability*.
  - **NOTE:** RabbitMQ doesn't support disaster recovery or installation between geographically distributed networks with high latency or poor link quality or deployments under cloud environments.
- **Genetec™ Incident Document Service:**
  - Installed as part of Mission Control.
  - It is an optional component that enables attaching links or documents to incidents in the *Incident monitoring* task.
  - Can be deployed on any Security Center server or expansion server.
  - Supports failover mode.
  - Incident Document Service role needs to be created to use the component.
  - For ease in maintenance, it is recommended to be deployed on the same server used for Mission Control.
- **SQL Server:**
  - The default database used by the Incident Manager role to store data.
  - Most SQL Server technologies support some form of clustering or mirroring.
- **Client tasks:**

  - On installation, Mission Control is integrated into the Config Tool and Security Desk clients.
- **Genetec Mission Control™ Web API:**
  - A suite of development tools that can be used by developers to create applications that interact with Mission Control.
  - Secure access to incident data.
  - Enables integration with other software systems.

Difference between high availability and failover

- **High availability:**
  - System can maintain operations for a high percentage of time even when failures arise.
  - Implies fault tolerance in this case. Fault tolerance is maintaining system operations without any downtime, despite machine or network failures.
  - RabbitMQ and SQL Server are the only components of Mission Control that support high availability.

  NOTE: High availability depends largely on your system infrastructure.
- **Failover:**
  - System incurs downtime when failures arise.
    The downtime isn't significant as it's the time taken to detect an offline role and turn on another instance on another machine.
  - Supports fault resilience. Fault resilience is the system's ability to detect failures and take appropriate actions to resume operations and bring back availability.

Key tip for Mission Control installation

Install the same version of Mission Control on all Security Center servers that need to host the Incident Manager role.

Integration with Sipelia

If you're planning to use Sipelia™ public addressing features, you must have the Sipelia Client installed on the Mission Control server.

Deployment options overview

You can deploy Genetec Mission Control™ in different ways depending on your existing system architecture and needs.

Determining the optimal deployment option for you

Each of the Components involved in Mission Control deployment on page 17 can have its own deployment option.

The options detailed here are per component basis. Each deployment option has its own benefits and drawbacks. If not deploying the basic option, it is recommended that you mix and match the deployment architecture option of each component to deploy a group of services on a single machine or multiple machines as required.

- Basic deployment option
- Security Center Directory deployment options
- Incident Manager role deployment options
- RabbitMQ deployment options
- SQL server deployment options

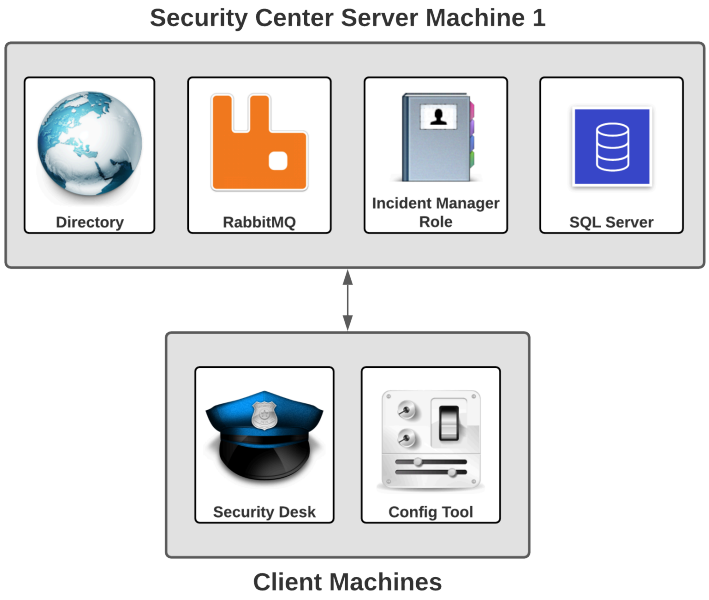
Basic deployment option

Simplest deployment architecture for Genetec Mission Control™ where all the services are running on the same machine.

Deployment architecture

The following services are configured to run on a single machine:

- Security Center Directory
- Mission Control Incident Manager role
- RabbitMQ server
- SQL Server



Benefits

- **Easy deployment:** You can execute all installers on one machine with no special configurations.
- **Low maintenance:** You only have one machine to monitor and maintain.
- **Low latency:** Services communicate directly with each other instead of going on the network.
- **Low impact from environment problems:** Network latency and failures never affect your deployment, unless they prevent Security Desk and Config Tool from accessing your machine. This does not affect services that would still operate normally.
- **Low cost:** As all components are on one machine the cost or deployment is relatively low.

Drawbacks

- **No fault resilience or fault tolerance:** Having your machine fail or go offline means you lose access to all services.
- **Downtime incurs data loss:** If all services are down, every event that triggers an incident in your system is lost forever. You would also lose incidents when the machine is down.
- **Updating your Windows machine might incur downtime:** Your system incurs a downtime during each update as it reboots.

NOTE: Cloud deployment is currently not supported for Mission Control.

Security Center Directory deployment options overview

You can deploy the Security Center Directory in various ways depending on your existing system infrastructure and business needs.

Security Center Directory deployment options

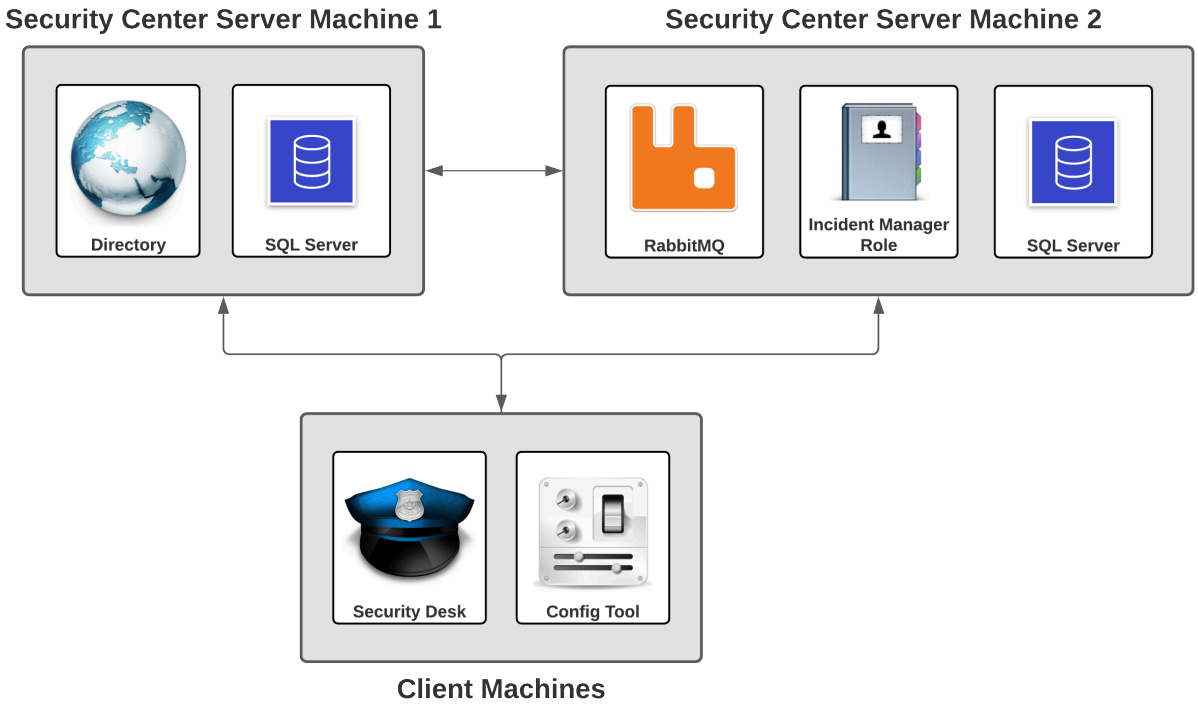
The Security Center Directory is vital component that logs all the events in your system. If it goes offline, all the events generated by entities such as cameras, door sensors, alarms, and so on, would be lost. In a basic deployment architecture, it is a significant drawback, as existing resources are shared.

To mitigate these issues, you can deploy the Security Center Directory in one of the following ways:

- **Option 1:** On a standalone machine
- **Option 2:** On same machine as other services with failover configuration on a standalone machine
- **Option 3:** On standalone machine with failover configuration on a separate standalone machine

Security Center Directory deployment option: standalone machine

You can deploy the Security Center Directory on a standalone machine separate from the rest of Genetec Mission Control™ components.



Benefits

- **Security Center Directory resources are not shared:** Other services do not use the same resources and therefore do not impact the Security Center Directory performance or availability.
- **Low maintenance:** Only two machines to monitor and manage; one for the Security Center Directory and one for all other Mission Control components.

Drawbacks

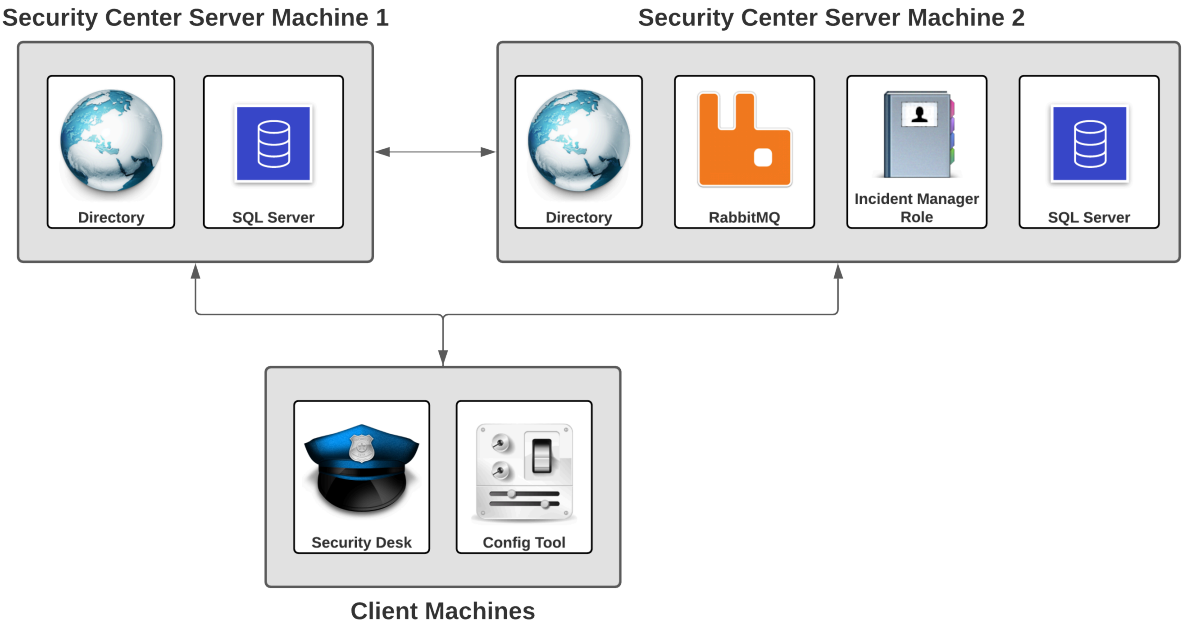
- **Complicated deployment:** This deployment model is slightly more involved as you would have to manually install the Security Center server on a separate machine and move the Security Center Directory to that server.
- **Higher latency in communications:** Communications between the Security Center Directory and all other Mission Control components might need a bit more time.
- **Network environment impacts up-time:** Issues in the network environment between the Security Center Directory and other Mission Control components directly impacts service.
- **No fault resilience or fault tolerance:** Issues with the machine on which the Security Center Directory is running impacts all other components. Losing connectivity to the machine with all the other services makes all those services unavailable, although the Security Center Directory would still be available.
- **Updating your Windows machine might incur downtime:** Every time you update your system, there is downtime incurred while the system reboots.

Security Center Directory deployment option: same machine as other services

You can deploy the Security Center Directory on the same machine as all the Genetec Mission Control™ components and have a failover configuration on a separate standalone machine.

What you should know

This deployment architecture is identical to the Basic deployment option on page 20 with an additional failover configuration for the Security Center Directory on a separate standalone machine.



For more information on configuration, see "Preparing Directory failover and load balancing" in the *Security Center Administrator Guide*.

This deployment architecture gives you the option of running your services in standard or failover mode.

Benefits

- **Low maintenance:** Only two machines to monitor and manage; one for the Security Center Directory and all other Mission Control components and one for the failover server.

- **Low latency:** Services communicate directly with each other instead of going on the network.
- **Low impact from environment problems:** Network latency and failures never affect your deployment unless they prevent Security Desk and Config Tool from accessing your machine. This does not affect services which will still operate normally.
- **Fault resilience for the Security Center Directory:** If the machine running the Directory goes offline, the system launches another instance of the Directory on another machine, with a short delay. This ensures continued operations with minimal disruptions.
- **Security Center Directory resources are not shared:** If the Directory is running on a standalone machine, other services do not use the same resources and therefore do not impact the Security Center Directory performance or availability unless failover is triggered.

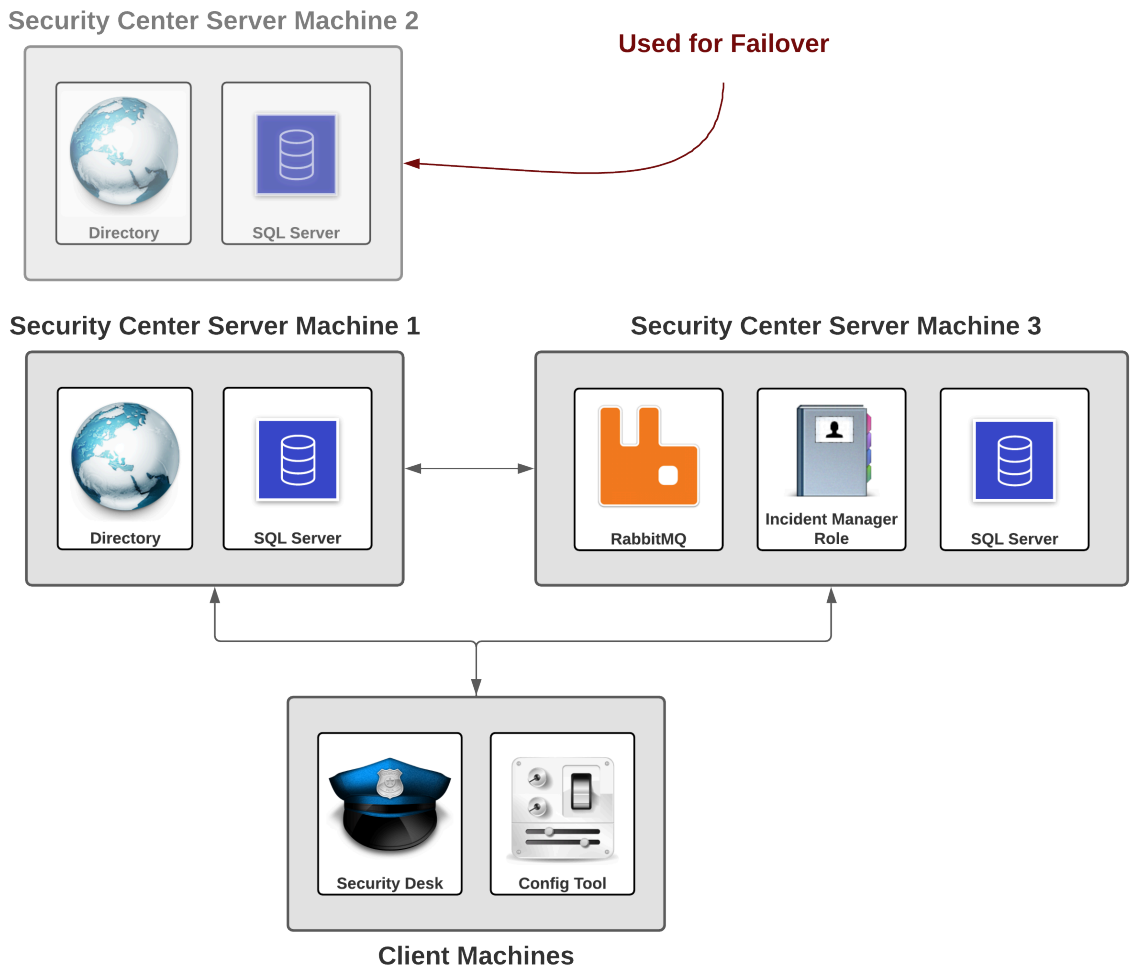
Drawbacks

- **Complex deployment:** You must install the Security Center server on a separate machine and manually add the Security Center Directory failover to that server.
- **No fault resilience or fault tolerance:** Having your machine with all the services fail or go offline means you lose Mission Control incident data. Directory remains available, unless both machines go offline.
- **Resource starvation in failover mode:** In failover mode, Directory and Incident Manager would be competing for the same resources.

Security Center Directory deployment option: Directory and failover on separate standalone machines

You can deploy the Security Center Directory on a standalone machine and configure Directory failover on another standalone machine.

You can isolate machine resources, even in failover mode, by using two machines with the Security Center Directory and its corresponding failover.



For more information on configuration, see "Preparing Directory failover and load balancing" in the *Security Center Administrator Guide*.

Benefits

- **Resource availability:** Resources are not shared even in failover mode.
- **Fault resilience for the Security Center Directory:** If the machine running the Directory goes offline, the system launches another instance of the Directory on another machine, with a short delay. This ensures continued operations with minimal disruptions.
- **Minimal downtime for rolling updates:** Updates for your Windows system are available with minimal downtime for the Security Center Directory.

Drawbacks

- **Medium maintenance:** With this deployment architecture, there are three machines to monitor and maintain.
- **Complex deployment:** You must install Security Center server on two machines, manually move the Security Center Directory to one machine and the Directory failover to the other machine.
- **Higher latency in communications:** There would be some delay in communications between the Security Center Directory and other components Genetec Mission Control™.
- **No fault resilience or fault tolerance:** Having your machine with all the services fail or go offline means you lose Mission Control incident data. Directory remains available, unless both machines go offline.

Quality of service

Using this deployment architecture implies that service quality relies more on your system infrastructure. The quality of service would be directly dependent on the quality and reliability of your network infrastructures.

Incident Manager deployment options overview

You can deploy the Incident Manager role in various ways depending on your existing system infrastructure and business needs.

Incident Manager role deployment options

The Incident Manager role is responsible for most of the Genetec Mission Control™ business logic. If the Incident Manager goes offline, RabbitMQ continues to register the missed events that the Incident Manager can use after it comes back online.

You can also leverage the Incident Manager role's ability to support failover.

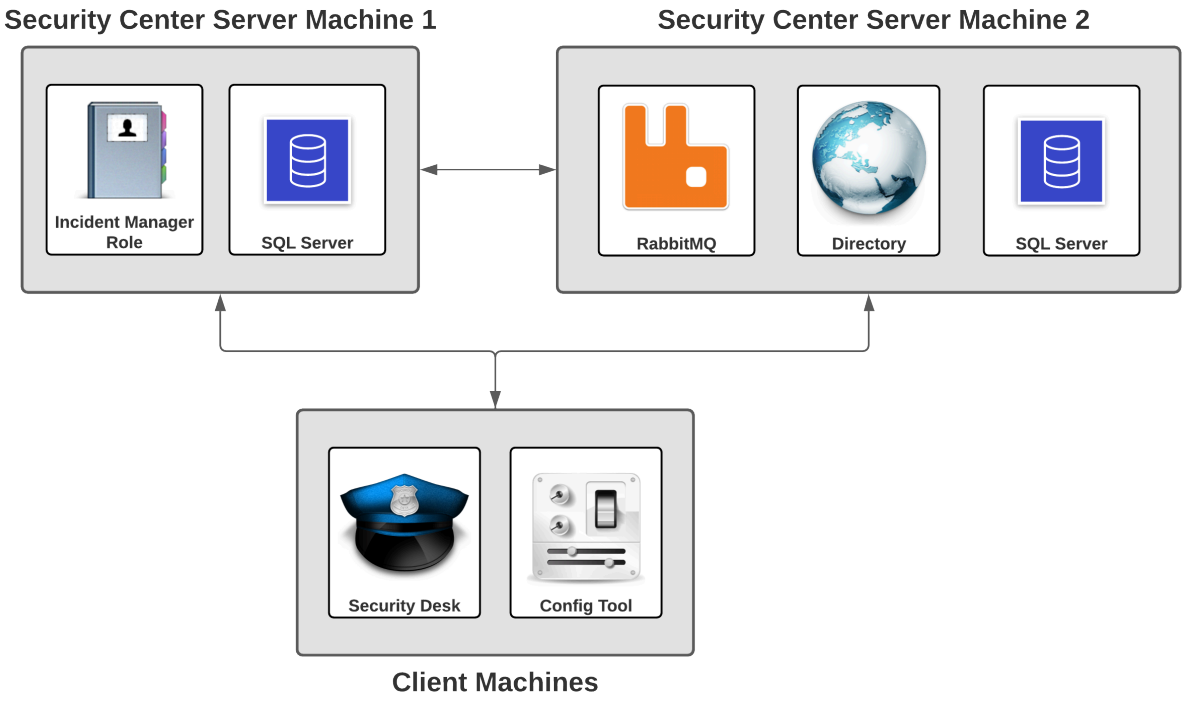
NOTE: It is not recommended to run the Incident Manager role on the Security Center Directory server.

You can deploy the Incident Manager role in one of the following ways to meet your business requirements:

- **Option 1:** On a standalone machine
- **Option 2:** On a standalone machine with Incident Manager failover configuration on the machine with the other services
- **Option 3:** On a standalone machine with failover on a separate standalone machine

Incident Manager deployment option: standalone machine

You can deploy the Incident Manager role on a standalone machine, separate from the other components of Genetec Mission Control™ installation.



Benefits

- **Incident Manager resources are not shared:** Other services do not use the same resources and therefore do not impact the Incident Manager's performance or availability.
- **Low maintenance:** Only two machines to monitor and manage; one for the Incident Manager role and one for all other Mission Control components.
- **Data loss tolerance for the Incident Manager role:** If the machine running the Incident Manager goes offline, Mission Control becomes unavailable. However, if RabbitMQ and the Security Center Directory are still online and running on other machines, no data is lost and the Incident Manager role triggers any missed incidents on becoming available.

  NOTE: If your downtime is significant, you would begin to notice a loss of data pertaining to incident triggers. Furthermore, the system takes time to assimilate the data accumulated during downtime, incurring a time delay.

Drawbacks

- **Complex deployment:** You must install the Security Center server or expansion server on a separate machine and manually move the Incident Manager role to that server.
- **Higher latency in communication between components:** There might be a slight delay in communications between the Incident Manager role and the other services.
- **Greater impact from environment issues:** Network environment issues between the Incident Manager role and other services would directly impact service up-time.
- **Single point of failure:** If the machine with the Security Center Directory and RabbitMQ goes offline, you would lose access to the whole system including all information on incidents, devices and entities.
- **Updating your Windows machine might incur downtime:** Every time you update your system, there is downtime incurred while the system reboots.

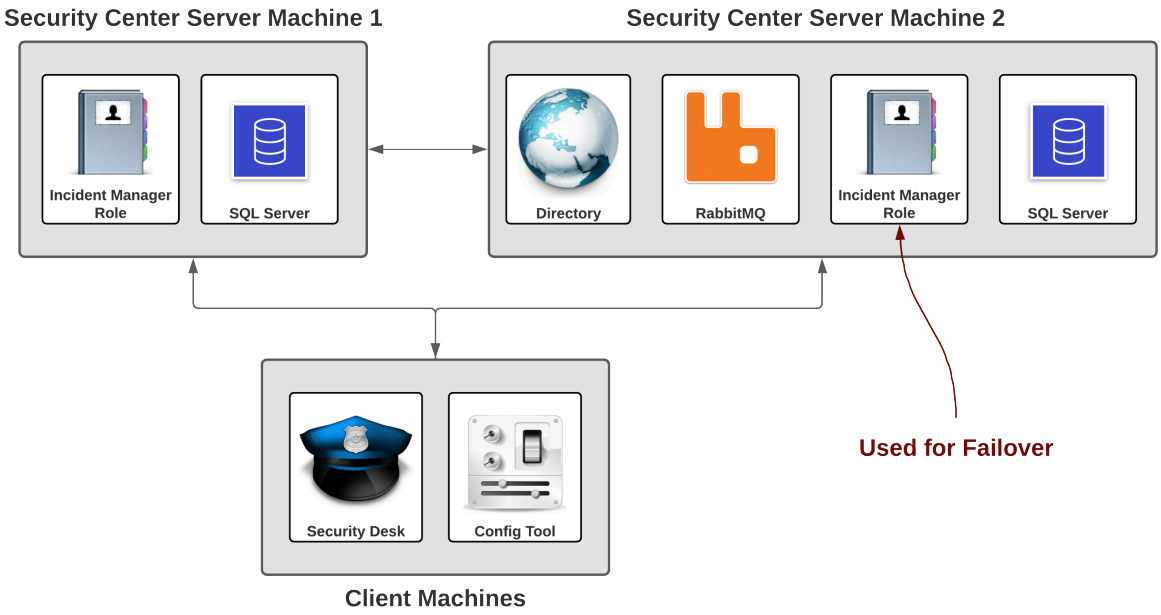
Quality of service

Using this deployment architecture implies that service quality relies more on your system infrastructure. The quality of service would be directly dependent on the quality and reliability of your network infrastructures.

Incident Manager deployment option: on standalone machine with separate failover configuration

You can deploy the Incident Manager on a standalone machine and configure failover for it on the machine with all other services.

The procedure to set up failover clusters for the Incident Manager role follows the same process as configuring role failover in Security Center. For more information, see Role failover and Setting up role failover.



Benefits

- **Low maintenance:** Only two machines to monitor and manage; one for the Incident Manager role and one for all other Genetec Mission Control™ components.
- **Advantages of failover configuration:** If the machine with the Incident Manager role goes offline, failover kicks in and you derive the same benefits of having the Incident Manager role running on the same machine as all other services.
- **Fault resilience for the Incident Manager role:** If the machine running the Incident Manager goes offline, the system launches another instance of the Incident Manager on the machine with the other services after a slight delay thereby resuming operations.

  NOTE: If however, any of the other services are down this option becomes useless as the Incident Manager requires all components to be up and running.
- **Incident Manager resources are not shared:** Other services do not use the same resources and therefore do not impact the Incident Manager performance or availability unless failover is triggered.
- **Minimal downtime for rolling updates:** Updates for your Windows system are available with minimal downtime for Incident Manager role.

Drawbacks

- **All drawbacks of deploying the Incident Manager role on a standalone machine:**
  - Complex deployment
  - Higher latency in communication between services
  - Greater impact from environment issues
  - Single point of failure
  - Downtime delay during Windows updates
- **No fault resilience or fault tolerance for other components:** Having your machine with all the services fail or go offline means you lose incident data. The Incident Manager role also becomes unavailable when any of the components go offline.

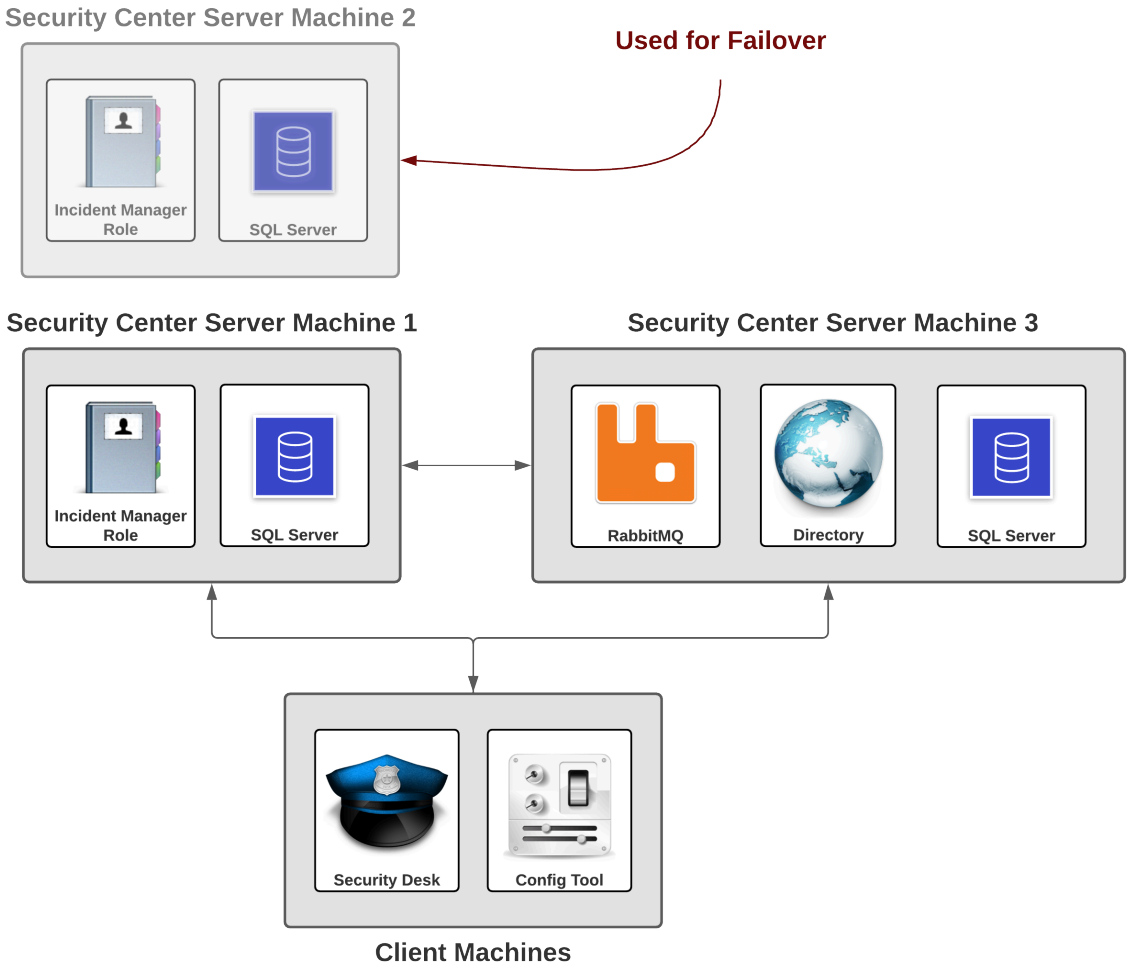
NOTE: For the Incident Manager deployment, you cannot have the failover on a standalone machine because all the other services are required for the Incident Manager to run smoothly. However, if all the

other components required for the Incident Manager to run support fault resilience or fault tolerance, you could try having a failover Incident Manager role on the same machine.

Incident Manager deployment option: Incident Manager and failover on separate machines

You can deploy the Incident Manager role on a standalone machine and configure failover on a separate standalone machine.

The procedure to set up failover clusters for the Incident Manager role follows the same process as configuring role failover in Security Center. For more information, see Role failover and Setting up role failover.



Benefits

- **Incident Manager resources are not shared:** Machine resources are not shared even in failover mode and therefore do not impact the Incident Manager performance or availability.
- **Fault resilience for the Incident Manager role:** If the machine running the Incident Manager goes offline, the system launches another instance of the Incident Manager on the machine with the other services after a slight delay thereby resuming operations.
- **Minimal downtime for rolling updates:** Updates for your Windows system are available with minimal downtime for Incident Manager role.

Drawbacks

- **Higher complexity in deployment:** You must install the Security Center server in two additional machines, move the Incident Manager role to one and configure Incident Manager failover in the other.
- **Medium maintenance effort:** There are three machines in this deployment architecture to monitor and maintain.
- **Higher latency in communication between components:** There might be a slight delay in communications between the Incident Manager role and the other services.
- **No fault resilience or fault tolerance for other components:** If any of the other components go offline, the lack of support for failover or high availability for them leads to data loss and service unavailability.

Quality of service

Using this deployment architecture implies that service quality relies more on your system infrastructure. The quality of service would be directly dependent on the quality and reliability of your network infrastructures.

SQL Server deployment options overview

You can deploy the SQL Server in various ways depending on your existing system infrastructure and business needs.

SQL Server deployment options

If you're using a licensed edition of SQL Server (like SQL Server Standard, SQL Server Business Intelligence, or SQL Server Enterprise), keep in mind that all databases are managed by Microsoft SQL Server in Security Center. By default, SQL Server is configured to use as much memory as is available on the system.

This could lead to memory issues if you're hosting SQL Server and many roles on the same server, especially on a virtual machine with little memory resources.

For more information on Security Center databases and networks, see Databases.

To protect the Directory database and avoid data loss, you can choose the below options to deploy:

- **Option 1:** Configuring SQL server cluster
- **Option 2:** Setting up Directory database failover
- **Option 3:** Separate databases for Security Center Directory and Incident Manager roles

SQL database requirements

The SQL Server Express edition included with Genetec Mission Control™ has a database size limit of 10 GB and doesn't support SQL clustering.

The average size of an active incident in the Incident Manager database is approximately 150 KB. If you have 3000 - 4000 incidents per day, you need 150 GB of database space with a configuration of 1-year retention period.

The average size of an incident type configuration is approximately 5 MB. The largest part of any incident type configuration is the **Automation** workflow configuration. Each time the incident type configuration is modified, a new version of the incident type is created and saved in the system. For example, if you have made 100 revisions to the incident type configuration, you need a minimum database space of 500 MB to save them.

If you need a longer retention period or require failover clustering for SQL Server high availability, it's recommended you use SQL Server Standard or Enterprise editions.

SQL Server database encryption

SQL Server offers the option to use Transparent Data Encryption (TDE) to encrypt database data files.

Mission Control uses a SQL database to operate and store files. To protect the database, configure TDE on the host SQL Server.

If TDE is used on the Mission Control database host SQL Server, the performance of the role decreases by 3%-5%.

SQL Server deployment option: clustering

The SQL database is essential for optimal operations. If the SQL database is lost all roles become unavailable. You can distribute your SQL database through clustering to prevent data loss.

To configure your SQL database server for a clustered environment, you must move the database to your mirrored data partition so that the contents of the database are identical on the active server and the standby server.

SQL Server deployment option: failover

The SQL database is essential for optimal operations. If the SQL database is lost all roles become unavailable. You can distribute your SQL database through failover configuration to prevent data loss.

Failover through backup

To protect the Directory database by regularly backing up the master database instance, you can set up Directory database failover using the backup and restore method.

For details, see the "Setting up Directory database failover through backup and restore" topic in the *Security Center Administrator Guide*.

Failover through mirroring

To protect the Directory database so you do not lose any data if a failover occurs, you can set up Directory database failover to use Microsoft SQL Server Database Mirroring.

CAUTION: Microsoft Database Mirroring is being phased out. For new installations, it is recommended to use SQL AlwaysOn.

For details, see the "Setting up Directory database failover through mirroring" topic in the *Security Center Administrator Guide*.

Failover through SQL AlwaysOn

If you are using the Windows feature **SQL AlwaysOn** as your Directory database failover solution, you must configure the Directory Manager to use SQL AlwaysOn in Config Tool.

For details, see the "Setting up Directory database failover through SQL AlwaysOn" topic in the *Security Center Administrator Guide*.

Reference resources

- For more details on database failover, see the "Directory database failover" topic in the *Security Center Administrator Guide*.
- For more details on creating high availability using SQL failover clusters, see SQL Server Server Failover Cluster Installation.
- For information on SQL Server permissions, see Assigning SQL permissions on remote SQL Servers on page 43.

SQL Server deployment option: independent of Security Center

The SQL database is essential for optimal operations. If the SQL database is lost all roles become unavailable. You can distribute your SQL database by installing it independent of Security Center to prevent data loss.

For details on installing SQL Server independently of Security Center, see the *Security Center Administrator Guide*.

RabbitMQ deployment options overview

You can deploy the RabbitMQ in various ways depending on your existing system infrastructure and business needs.

RabbitMQ deployment options

RabbitMQ is a sensitive component, as loss of the RabbitMQ service leads to data loss and downtime for all services except the Security Center Directory.

On the plus side, RabbitMQ supports the highest tolerance, albeit the cost of resources and deployment complexity. To leverage this feature, you must have a minimum of three standalone machines. While adding additional nodes to the RabbitMQ cluster improves fault tolerance, it does not add to performance.

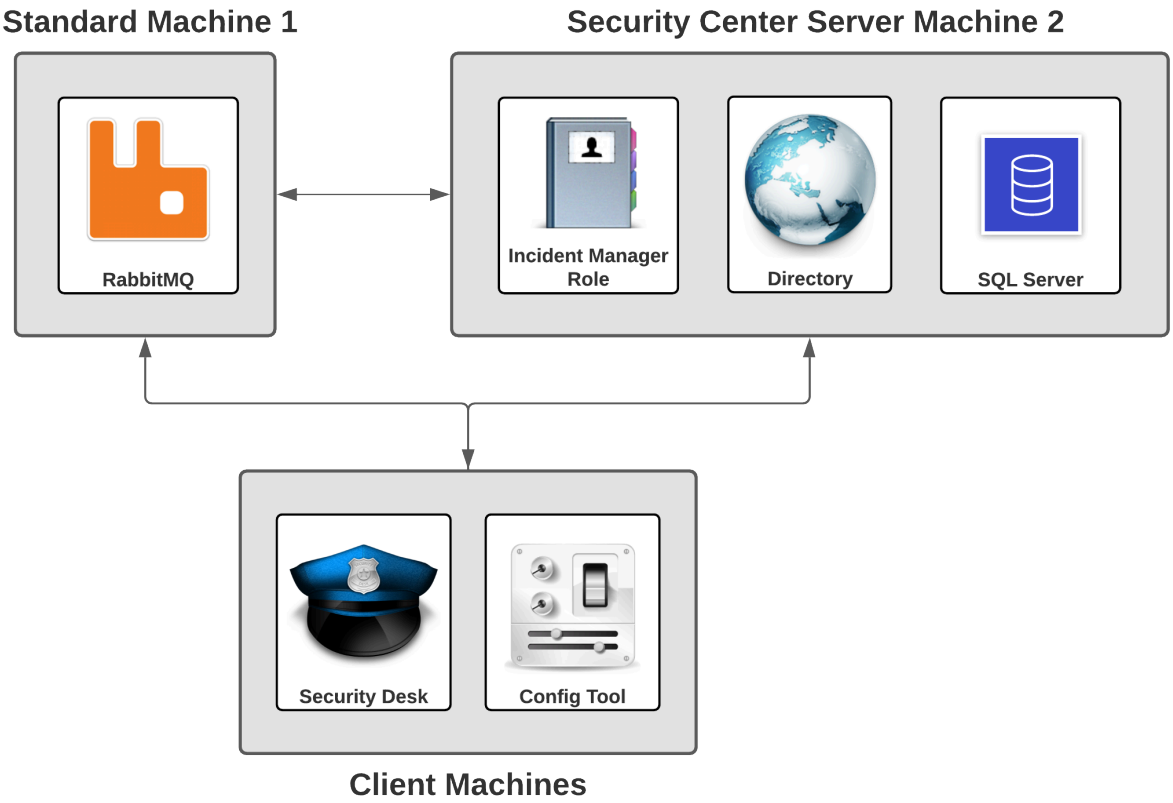
You can deploy RabbitMQ in one of the following ways depending on your business needs:

- **Option 1:** On a standalone machine
- **Option 2:** As a standalone cluster on three or more machines

NOTE: If RabbitMQ is not installed on the same server as the Incident Manager role, copy the RabbitMQ certificates to the Incident Manager server.

RabbitMQ deployment option: standalone machine

You can deploy RabbitMQ on a standalone machine separate from all other components of the Genetec Mission Control™ installation.



Benefits

- **RabbitMQ resources are not shared:** Other services do not use the same resources and therefore do not impact RabbitMQ performance or availability.
- **Low maintenance:** Only two machines to monitor and manage; one for RabbitMQ and one for all other Mission Control components.

Drawbacks

- **Complex deployment:** You must manually execute the Genetec Mission Control installer on the second machine, clear all component installation options and select only the **RabbitMQ** option.
- **Higher latency in communication between components:** There might be a slight delay in communications between RabbitMQ and the other services.
- **Greater impact from environment issues:** Network environment issues between RabbitMQ and other services would directly impact service up-time.
- **No fault resilience or fault tolerance:** If the machine with RabbitMQ goes offline you would lose the Incident Manager role as well along with all the data on incident triggers.

  If the machine with the other components goes offline you would lose availability of all services as the Security Center Directory would be offline as well.

  You can resolve Security Center Directory downtime issues by employing a failover configuration for the Security Center Directory:

  - Security Center Directory on the same machine as other services with failover configuration on a standalone machine.
  - Security Center Directory on a standalone machine with failover configuration on a separate machine.
- **Updating your Windows machine might incur downtime:** Every time you update your system, there is downtime incurred while the system reboots.

Quality of service

Using this deployment architecture implies that service quality relies more on your system infrastructure. The quality of service would be directly dependent on the quality and reliability of your network infrastructures.

RabbitMQ deployment option: standalone clusters

You can deploy RabbitMQ in a cluster configuration on a minimum of three standalone machines.

What you should know

This deployment architecture requires a minimum of three standalone machines (nodes). They must all be on the same local area network (LAN).

NOTE: Cloud deployment is currently not supported for Genetec Mission Control™.

For this method of deployment to work the following requirements must be met:

- The network between the nodes should be reliable, have a high performance threshold, and have low latency.
- If deploying nodes in separate geographical regions, you must be able to guarantee the above metrics. For example, most cloud providers have very reliable links between their data centers.
- The nodes should be on different server racks for better tolerance if deployed on-premises.
- While more nodes improve resilience, they do not improve performance.

It is recommended that you have thorough knowledge of RabbitMQ configurations. For details, see the official RabbitMQ documentation.

High availability in RabbitMQ clusters

In a three-node cluster, losing one node even when the other two nodes are running causes a degradation in performance.

The high availability configuration of RabbitMQ uses the Raft Consensus algorithm which involves multiple servers agreeing on values.

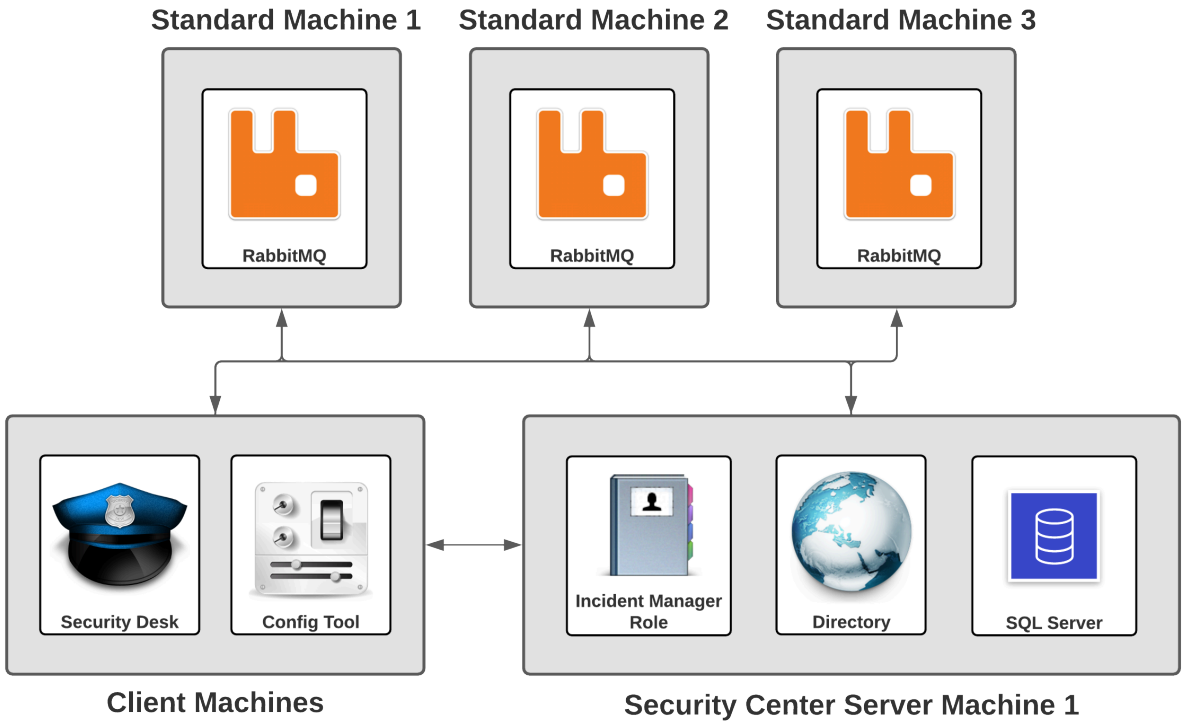
In this scenario, this translates to having more nodes active than offline for effective operations. The formula to calculate the number of nodes you would need for effective system availability is `(N/2)+1`, where `N` is the number of nodes.

Active RabbitMQ cluster scenarios

Number of nodes in cluster	Number of nodes that must be available	Number of node failures supported
1	1	0
2	2	0
3	2	1
4	3	1
5	3	2
6	4	2
7	4	3

NOTE: It is imperative that all the nodes have an active link for communication between them for the Raft Consensus algorithm and therefore your cluster to work.

The bigger the cluster, the more impact it has on performance. When nodes fail and the cluster is performing in a degraded state, the performance would be greatly reduced.



To set up high availability for RabbitMQ, see Configuring high availability for RabbitMQ on page 39.

Benefits

- **High availability:** Interruption to operations is minimal, even if a node is down.
- **RabbitMQ resources are not shared:** Other services do not use the same resources and therefore do not impact RabbitMQ performance or availability.
- **No downtime for rolling updates:** You can update the RabbitMQ setup one node at a time without affecting service availability.

Drawbacks

- **Highly complex deployment:** RabbitMQ must be deployed on a minimum of three nodes. The deployment process is long and involved.
- **High maintenance:** In the rare event that a whole cluster goes offline because of multiple network partitions, you must manually fix the issue. It is recommended to have good knowledge of RabbitMQ configurations and diagnostics.

  For details, see the official RabbitMQ documentation.

Quality of service

Using this deployment architecture implies that service quality relies more on your system infrastructure. The quality of service would be directly dependent on the quality and reliability of your network infrastructures.

Deploying RabbitMQ on a separate server

If RabbitMQ is not deployed on the same server as the Incident Manager role, you must copy a certificate from the RabbitMQ server to the server hosting the Incident Manager role after installing or upgrading RabbitMQ.

Before you begin

Install or upgrade RabbitMQ on a separate server from the Incident Manager role. For more information, see Installing Genetec Mission Control™.

Procedure

1 On the server hosting the Incident Manager role, create the following folder: *C:\Program Files (x86)\Genetec\RabbitMQ*

2 On the server where RabbitMQ is installed, navigate to *C:\Program Files (x86)\Genetec\RabbitMQ* and copy the certificate called *Cert.pem*.

3 On the server hosting the Incident Manager role, navigate to *C:\Program Files (x86)\Genetec\RabbitMQ* and paste the *Cert.pem* certificate.

4 Repeat steps 2-3 for each server that has RabbitMQ installed separately from the Incident Manager role.

IMPORTANT: If you are copying multiple certificates from separate RabbitMQ servers, you must rename them. For example, *Cert_server1.pem* and *Cert_server2.pem*. Renamed certificates must not contain *CA* or *KEY*.

5 Repeat steps 1-4 for all Security Center servers hosting the Incident Manager role.

6 Restart the Incident Manager role.

Configuring high availability for RabbitMQ

To set up a resilient RabbitMQ cluster with high availability, you must set up at least three nodes. If any one of the nodes becomes unavailable, two of the remaining nodes can be reached and no data is lost.

What you should know

Nodes can be physical or virtual machines. All machines in the RabbitMQ cluster must be reachable. It is recommended that all machines in the RabbitMQ cluster be on the same domain and on the same local area network (LAN).

Procedure

1 Deactivate the Incident Manager role.

2 Use the Genetec Mission Control™ installer to install RabbitMQ on each machine where you need to create a node.

For instructions, see Installing Mission Control.

3 Create a RabbitMQ cluster but do not connect it to the Incident Manager role at this stage.

4 Add nodes to the RabbitMQ cluster.

5 For Mission Control versions before 3.0.2.0, enable queue mirroring on each node so they become highly available.

NOTE: From Mission Control 3.0.2.0 onwards, queue mirroring is handled by the Incident Manager.

6 Activate the Incident Manager role.

Creating a RabbitMQ Cluster

A RabbitMQ cluster is a feature that enables the synchronization of state data and configurations of multiple nodes in a real-time networked environment.

Procedure

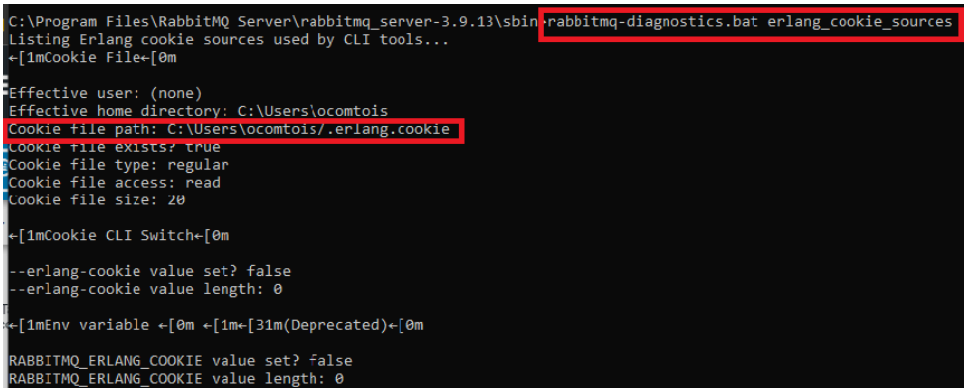
To create a RabbitMQ cluster:

1 Copy the *.erlang.cookie* files from the node 1 by following the below steps:

a. Open the RabbitMQ Command Prompt (sbin dir) on the machine that you want to copy the *erlang_cookie*.

b. Run the following command:

```
rabbitmq-diagnostics.bat erlang_cookie_sources
```



c. Take note of the *erlang.cookie* file path.

d. Copy the *erlang.cookie* from that file path.

e. In your Windows system, overwrite the *erlang.cookie* file in the below locations on node 1:

- %USERPROFILE%

  This usually leads to C:\Windows\ServiceProfiles\LocalService\ for a service
- %HOMEDRIVE%%HOMEPATH%

  This usually leads to C:\Users\<username >
- C:\
- C:\Windows\System32\config\systemprofile

NOTE: Use the same cookie file path for all destination nodes but ensure that the same *erlang.cookie* is overwritten on all locations of each node.

2 Copy the following certificate files from the node 1 and overwrite them to the other nodes:

- *CACert.pem*
- *Cert.pem*
- *Key.pem*
- *CAKey.pem*

By default, these files are found in *C:\Program Files (x86)\Genetec\RabbitMQ*.

NOTE: If the Incident Manager role is on a separate machine, then you must copy the certificates from the RabbitMQ server to the server hosting the Incident Manager role.

Modify the RabbitMQ configuration files on the node 1 to enable partition handling:

1 Right-click the Notepad application, select **Run as administrator**, and then open the *RabbitMQ.conf* file in Notepad.

By default, the RabbitMQ configuration file *RabbitMQ.conf*, is found in *C:\ProgramData\Genetec\RabbitMQ \config*.

TIP: Using Notepad++ with Makefile set as a language improves the readability of the file.

2 Modify the following line from:

```
# cluster_partition_handling = pause_minority
```

to:

```
cluster_partition_handling = pause_minority
```

By removing the `#` symbol, the line is no longer flagged as a comment and becomes part of the code.

3 Save the file.

Reinstall the RabbitMQ service to enable the configuration file changes:

1 Using the Windows search box, find and open the RabbitMQ Command Prompt (sbin dir).

2 Right-click and select **Run as administrator**.
This is installed with RabbitMQ by default.

3 Run the following commands:

a) `rabbitmq-service.bat remove`
When successfully run, the following message is displayed:

```
Service RabbitMQ removed from system
```

b) `rabbitmq-service.bat install`
When successfully run, the following message is displayed:

```
Service RabbitMQ added to system
```

c) `rabbitmq-service.bat start`
When successfully run, the following message is displayed:

```
The RabbitMQ service is starting. The RabbitMQ service was started successfully.
```

Ensure that your modified configuration file is loaded in memory:

1 In the RabbitMQ Command Prompt application, run the following command: `rabbitmqctl.bat environment`
When successfully run, the content of the *RabbitMQ.conf* file is loaded in memory and the following message is displayed in a format similar to a *JSON* file:

```
Application environment of node rabbit@<User>
```

2 Verify that your modifications have been implemented correctly:

a) Ensure that the modified configuration reads as follows:

```
cluster_partition_handling = pause_minority
```

b) If the line of code doesn't read as in the previous step, modify it to match.

Copy the modified *RabbitMQ.conf* file to the other nodes:

1 Copy the modified *RabbitMQ.conf* file from the node 1.
The default location of the config file is *C:\ProgramData\Genetec\RabbitMQ\config*.

2 Overwrite the existing *RabbitMQ.conf* file in the same file path of the destination nodes.
It's recommended to do a backup before you overwrite the existing files.

3 Reinstall RabbitMQ service for each node.

After you finish

Refer the recommended configurations and best practices for using virtual machines with RabbitMQ clusters.

Adding nodes to RabbitMQ cluster

Add backup nodes to the RabbitMQ cluster to help you enable queue mirroring. This ensures that messages published to a queue are replicated to all mirrors.

Before you begin

You must open port 25672 on every node.

Procedure

1 In Windows search box, type navigate RabbitMQ Command Prompt (sbin dir).

2 Right-click and select **Run as administrator**.
This application is installed with RabbitMQ by default

3 To stop the node, execute the following command:
`rabbitmqctl.bat stop_app`

4 To add the node to the cluster, execute the following command in the backup node:
`rabbitmqctl.bat join_cluster rabbit@<NodeName>`
NOTE: *<NodeName>* is case sensitive.

5 To restart the node, execute the following command:
`rabbitmqctl.bat start_app`
NOTE: You can verify that the correct node has been added successfully by executing the following command:

`rabbitmqctl.bat cluster_status`

or through the RabbitMQ web interface at

https://<node hostname>:15671

6 In Windows, open the *Services* application and restart the RabbitMQ service on all servers.

After you finish

Configure RabbitMQ servers in the Incident Manager role. See the *Genetec Mission Control™ Administrator Guide* for more details.

Related Topics

Ports used by applications in Mission Control on page 10

Assigning SQL permissions on remote SQL Servers

You must have the necessary SQL permissions on the remote SQL server with Genetec™ Server services where the Incident Manager role is active.

Before you begin

Install SQL Server Management Studio (SSMS) on your system. You can download an instance from Microsoft.com.

For information on installing SQL Server on a separate drive, see the *Security Center Installation and Upgrade Guide*.

What you should know

Security Center offers two modes of user authentication whilst connecting to the SQL server:

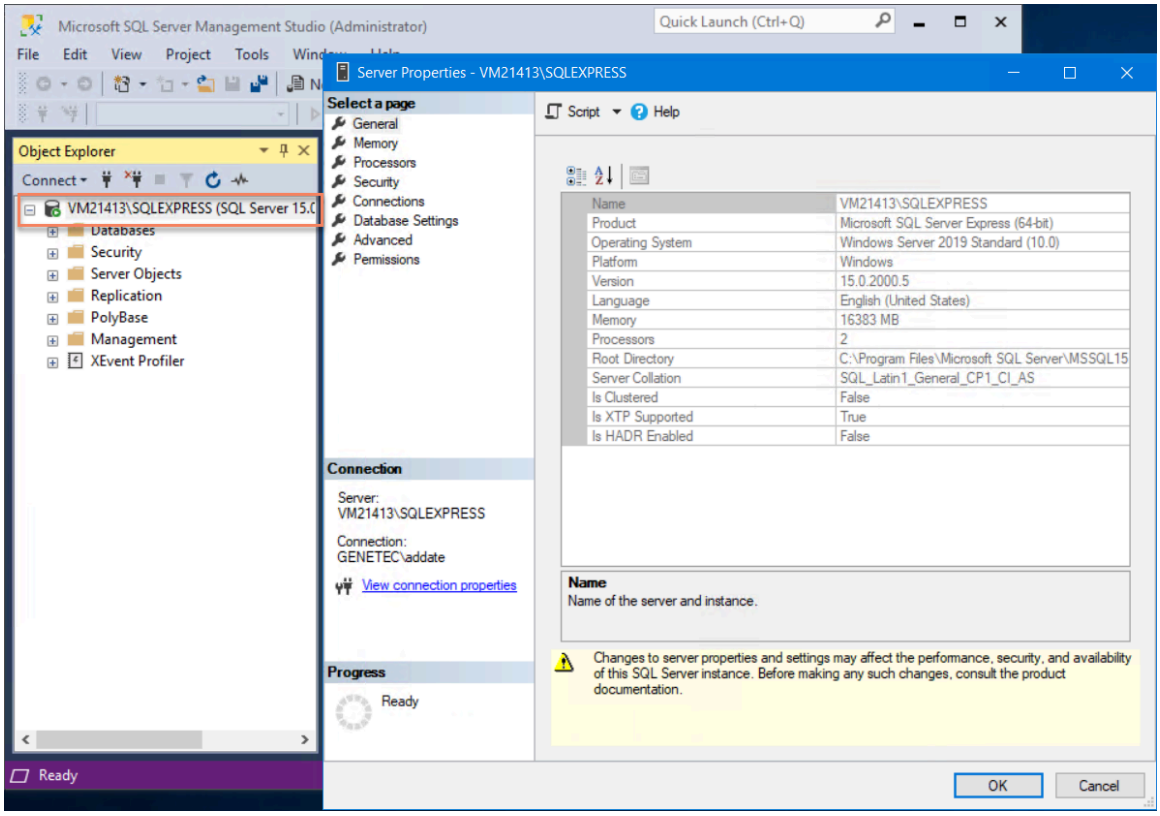
- **Windows Authentication:** Configured by default. The SQL server has the same login credentials as the Windows administrator.
- **SQL Server Authentication:** Configure for advanced scenarios. This enables dual authentication because when you configure this option, the default **Windows Authentication** is also available.

Procedure

To configure dual authentication:

1 Open SQL Server Management Studio.

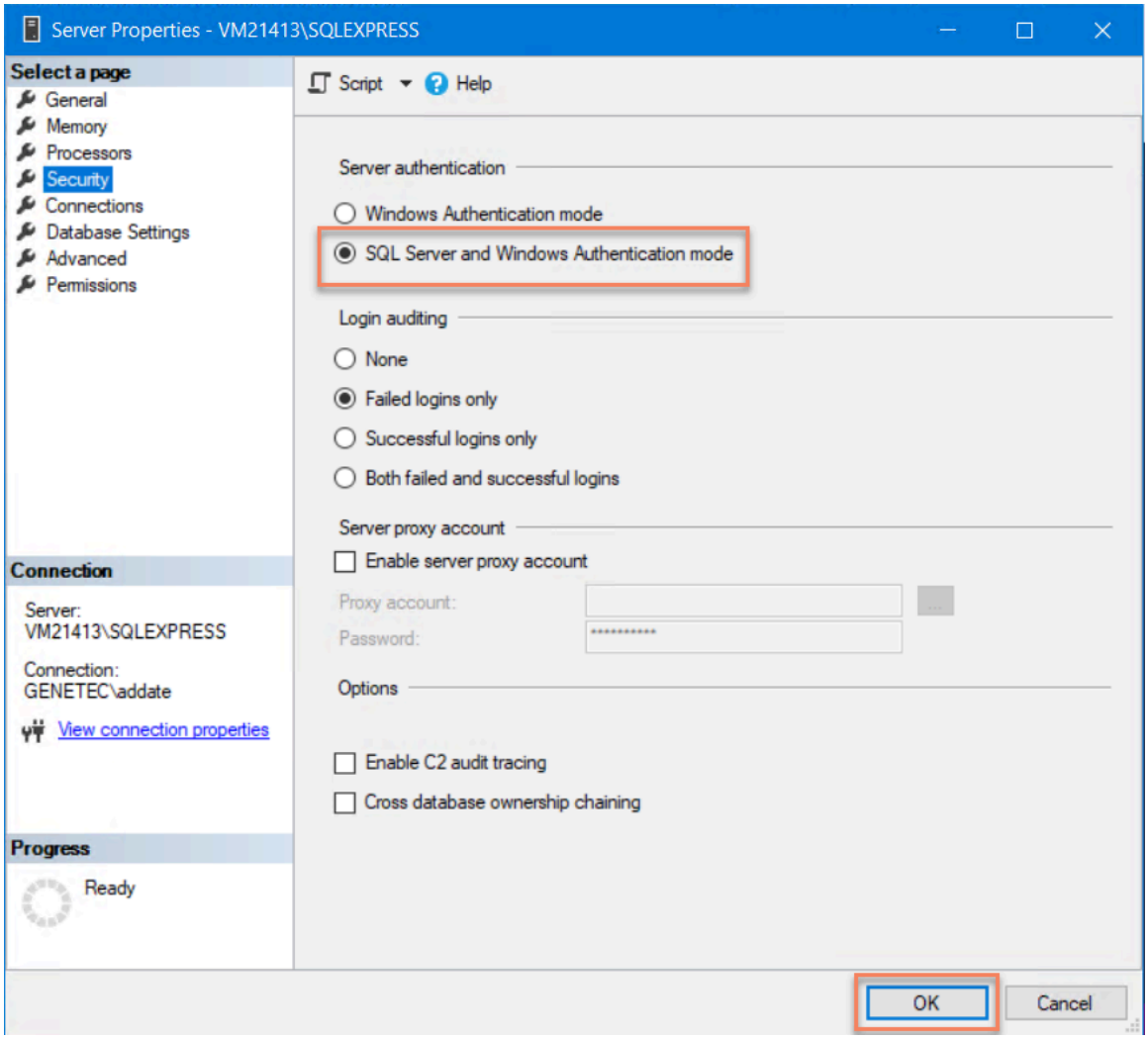
2 In the *Object Explorer* pane, right-click the SQL Server instance and select **Properties**.



The *Server Properties* dialog box opens.

3 In the *Select a page* pane, click **Security**.

4 In the *Server authentication* section, select **SQL Server and Windows Authentication mode** and click **OK**.



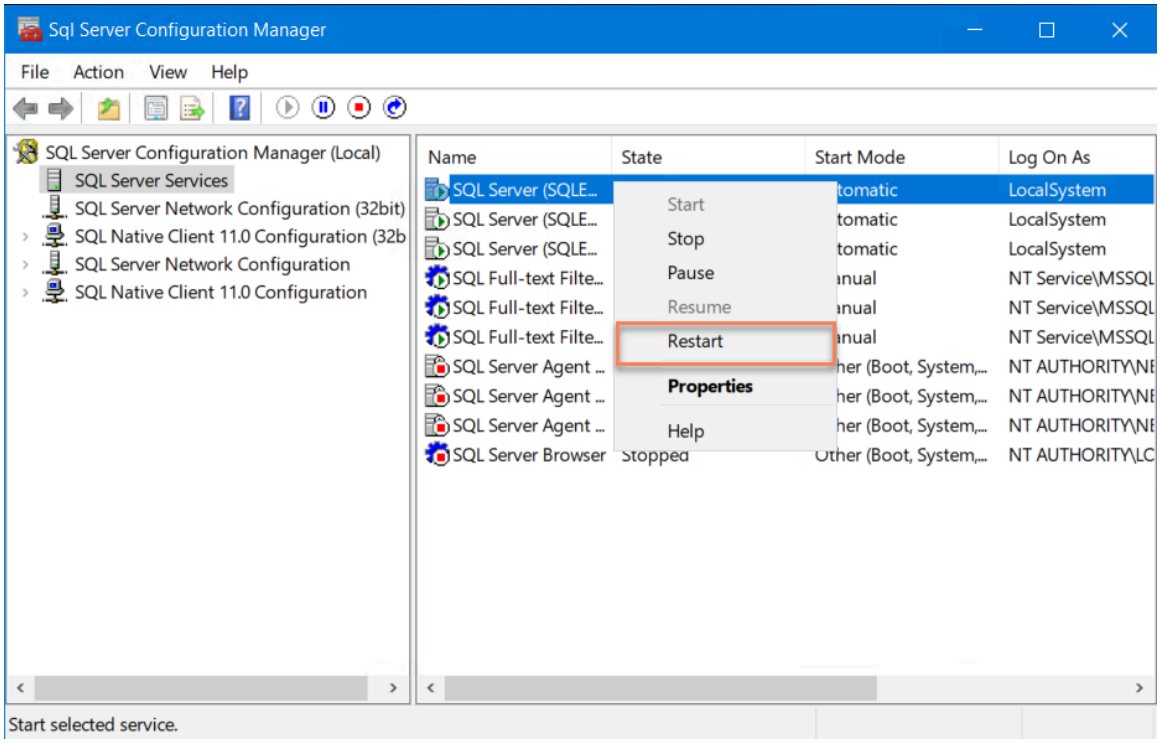
NOTE: For your configuration to take effect, you must restart your SQL Server.

5 Restart SQL Server:

a) Open SQL Server Configuration Manager.

b) In the *Sql Server Configuration Manager* pane, select **SQL Server Services**.

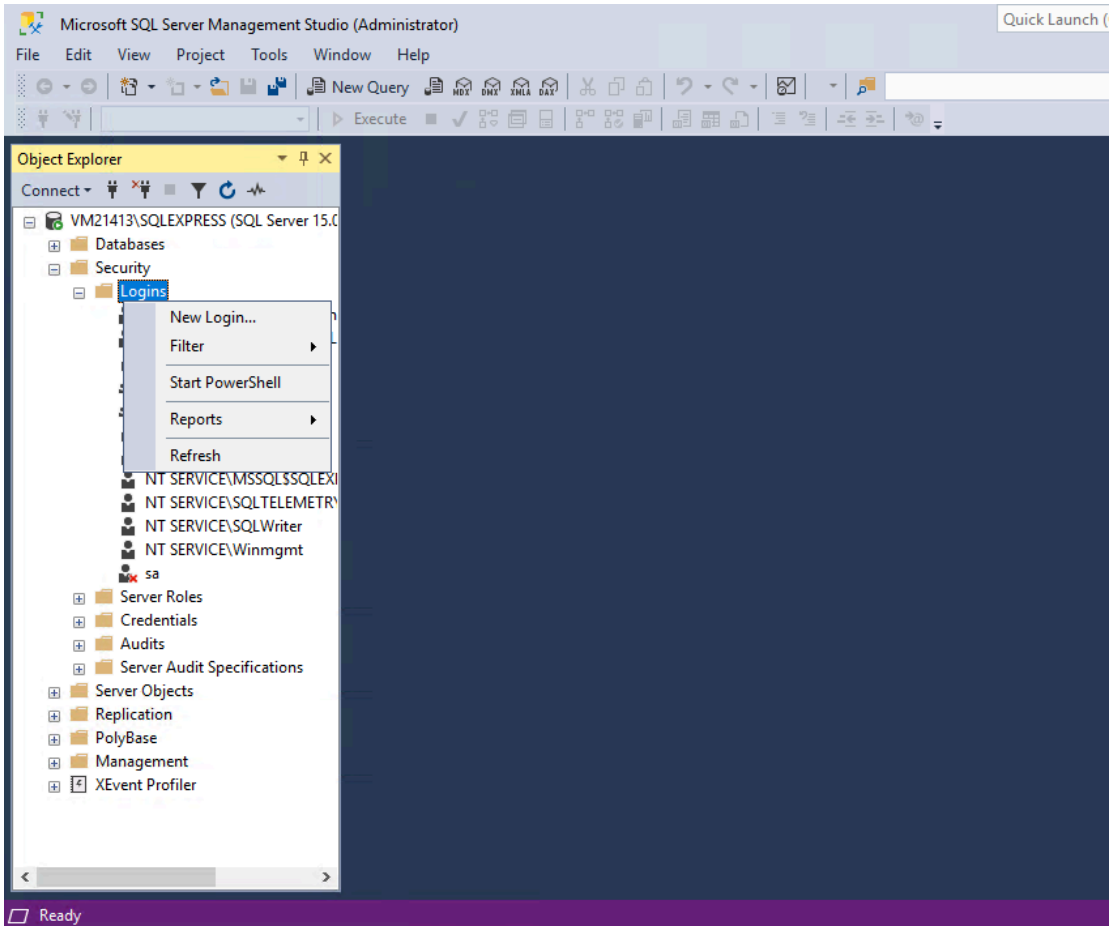
c) Right-click **SQL Server (SQLEXPRESS)** and click **Restart**.



Dual authentication is now configured.

6 Create a new SQL Server login using **SQL Server Authentication**:

a) In the *Object Explorer* pane, click **Security** > **Logins** > **New Login**.



The *Login- New* window opens.

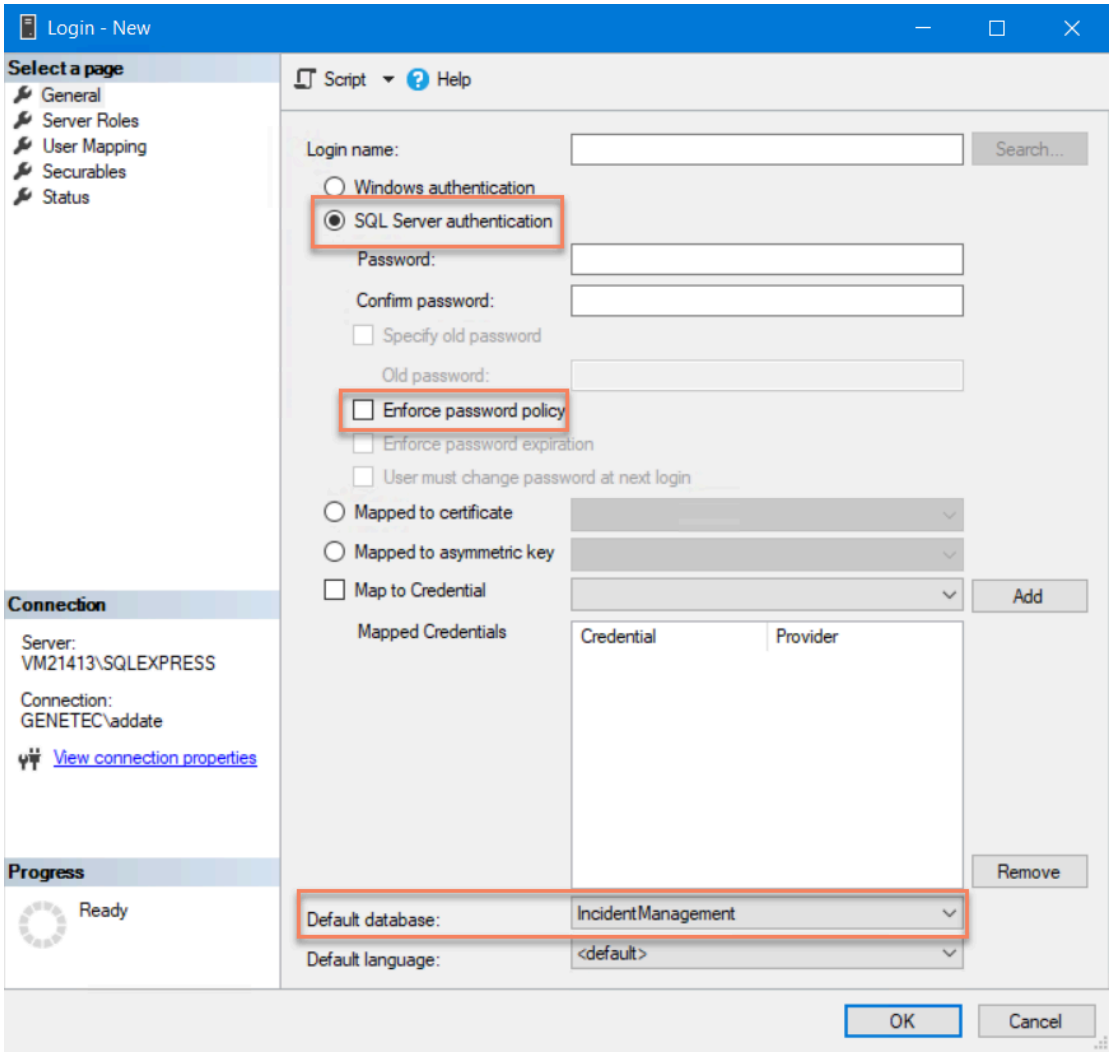
b) Select **SQL Server Authentication**.

TIP: Remember the login credentials for your SQL server. You would require Administrator level access to reset them.

c) Enter a login name.

d) Enter a new password.

e) Clear the **Enforce password policy** option.

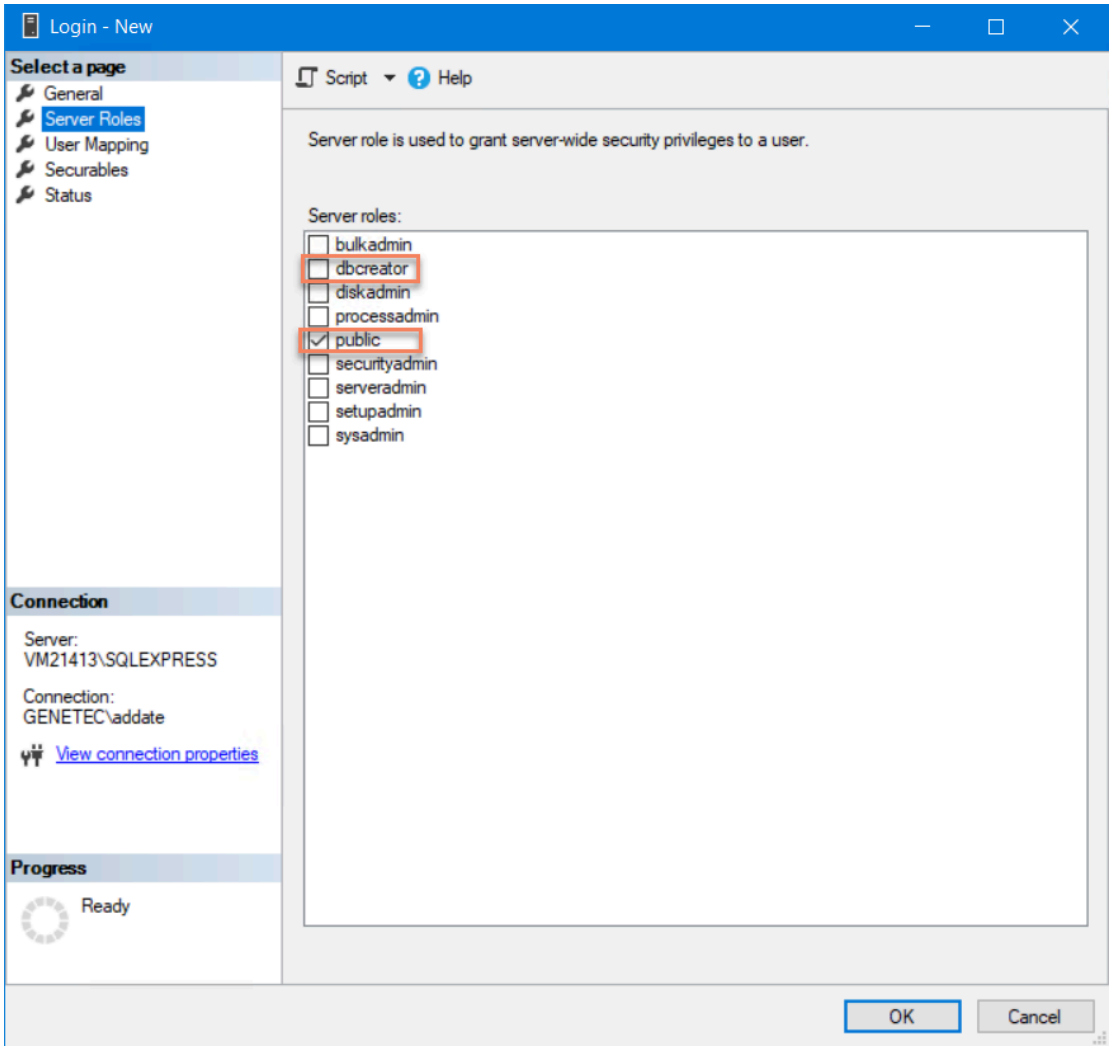


f) From the **Default database** list, select **IncidentManagement**.

g) In the *Select a page* pane, select **Server Roles**.

h) In the *Server Roles* section, select one of the following:

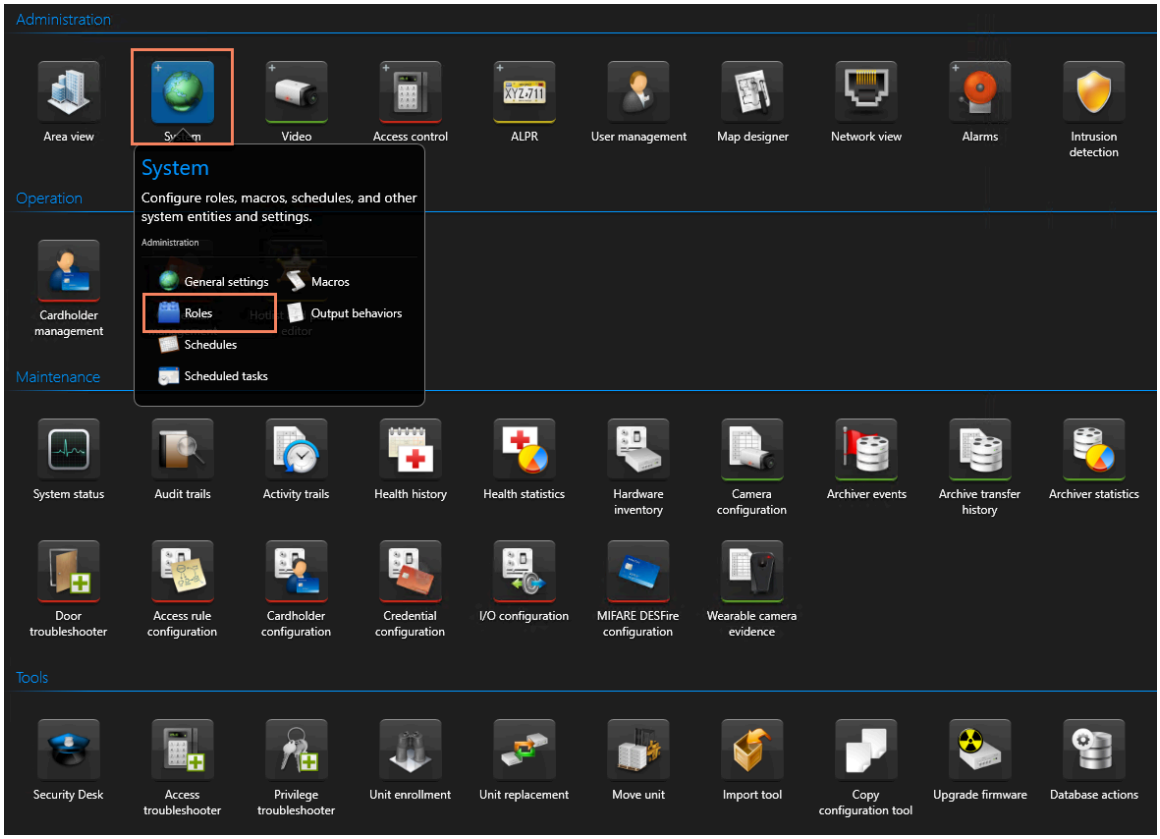
- **public:** For Security Center versions 5.10.1.0 and later.
- **dbcreator:** For Security Center versions 5.10.0.0 and earlier.



i) Click **OK**.
A new SQL Server login is created.

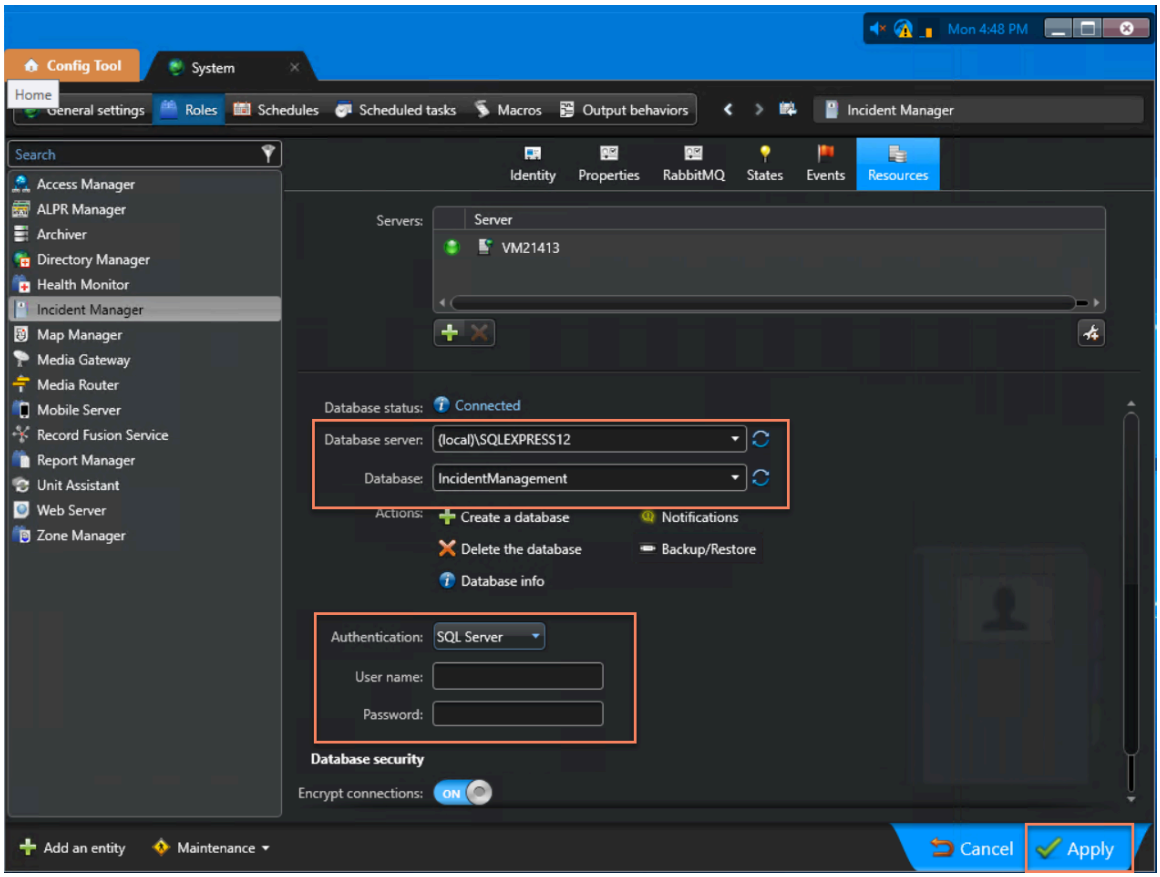
To test the new SQL Server login:

1 In Config Tool, go to **System** > **Roles** > **Incident Manager**.



2 On the *Resources* page, configure the following:

- **Database server and Database:** The name for the **Database server** and **Database** must match the name found at **Security** > **Login** > **New Login** > **Default database** name.
- **SQL Server login:** From the **Authentication** list, select **SQL Server** and enter the new SQL Server credentials that you created.



3 Click **Apply**.

The database is connected to the server with the new SQL server credentials.

Determining the best deployment architecture for you

You can mix and match the deployment options of the various components of Genetec Mission Control™ to build the best system architecture to suit your existing infrastructure and business needs.

What you should know

- Each of the Components involved in Mission Control deployment on page 17 can have its own deployment option.
- All the components can be installed on one machine as in the Basic deployment option on page 20.

  NOTE: If you choose this deployment architecture, ensure that your machine is high performing and has a high degree of fault tolerance.
- When deploying RabbitMQ:
  - RabbitMQ clusters must always be on the same local area network (LAN).
    - It should not be geographically distributed.
    - Cloud deployments are not supported.
    - There should be a high performance network between the nodes.
    - **NOTE:** RabbitMQ does not fully support disaster recovery as it requires asynchronous replication of data between data centers.
  - Adding more nodes to a RabbitMQ cluster improves resilience but does not improve performance.

4

Installing Mission Control

This section includes the following topics:

- "Mission Control installation overview" on page 54
- "Mission Control privileges" on page 75
- "Uninstalling Mission Control " on page 78
- "Mission Control Web API and UI SDK" on page 80

Mission Control installation overview

There are many tasks involved in installing Genetec Mission Control™. Use this overview as a reference to guide you through the process.

The following table lists the steps required to deploy Mission Control.

Step	Description
Prerequisites	
Read the release notes, supported upgrade paths, system requirements, and other information about the release.	• Release Notes. • Mission Control 3.4.1.0 system requirements on page 7. • Supported upgrade paths to Mission Control 3.4.1.0 on page 87.
Close all applications related to Microsoft Management Console (MMC), such as Services, Event Viewer, and so on. They might lock the Security Center services and prevent them from being updated.	
Install a supported version of Security Center.	Product compatibility for Mission Control 3.4.1.0 on page 13.
Verify that the Directory is connected to the internet.	For systems where the Directory is not connected to the internet, you need to disable Windows automatic root certificate updates.
Install Mission Control	
Download Mission Control	1. Visit the Product Download page in the Genetec Portal. **NOTE:** You need a username and password to log on to the Genetec Portal. 2. From the **Download Finder** list, select your version of Security Center. 3. From the **Mission Control** section, download the required Mission Control installation package. You can also get the Mission Control installation package from your Genetec™ Inc. representative. 4. Extract the package and double-click *setup.exe*.
Install Mission Control **IMPORTANT:** Install the same version of Mission Control on all Security Center servers that need to host the Incident Manager role.	Installing Mission Control 3.4.1.0 on page 56. / Installing the Mission Control patch release on page 68. Installing Mission Control in silent mode on page 69. Choose setup language. Review the License Agreement and accept the terms. Select the program features you want to install. • Mission Control • Genetec™ Incident Document Service • RabbitMQ Server Custom setup is also useful when you want to install a specific program on another server for high availability. **NOTE:** Genetec Mission Control™ Web API is not part of the selection as it is installed by default.

Step	Description
	Configure Genetec Mission Control Web API to access a suite of development tools. These tools can be used to create applications that interact with Mission Control and have secure access to incident data when integrated with other software systems.
	Configure the Genetec Incident Document Service to enable attaching links or documents to incidents.
	Configure RabbitMQ to communicate between the various components of Mission Control. You can either choose to automatically generate a self-signed RabbitMQ certificate or use a custom certificate. Default RabbitMQ ports: • SSL = 5671 • HTTPS API = 15671 **BEST PRACTICE:** Ensure optimal performance of the RabbitMQ service and avoid potential issues by configuring your anti-virus software to exclude the RabbitMQ database folder during scans.
What to do next	
In Config Tool, create the Incident Manager role.	Creating the Incident Manager role on page 71.
In Config Tool, create the Incident Document Service role.	Creating the Incident Document Service role on page 73
Grant users the privileges needed to use Mission Control.	Mission Control privileges on page 75.

Disabling Windows automatic root certificate updates

Windows systems that are not connected to the internet face connection issues when automatic root certificate updates are not turned off.

What you should know

This procedure is required for all systems where the Directory isn't connected to the internet and also on all computers running a Directory role.

Procedure

1 Open Command Prompt as an administrator and run *gpedit.msc*.

2 Go to **Local Computer Policy** > **Computer Configuration** > **Administrative Templates** > **System** > **Internet Communication Management** > **Internet Communication Settings**.

3 Set **Turn off automatic root certificate update** to **Enabled** and click **Apply**.

4 Go to **Local Computer Policy** > **Computer Configuration** > **Windows Settings** > **Security Settings** > **Public Key Policies** > **Certificate Path Validation Settings**.

The Certificate Path Validation Settings Properties dialog opens.

5 In the **Network Retrieval** tab select **Define these policy settings**.

6 Deselect **Automatically update certificates in the Microsoft Root Certificate Program(recommended)**.

7 In the **Default retrieval timeout settings** section, set the timeout values for **Default URL retrieval timeout** and **Default path validation cumulative retrieval timeout** to 1 second and click **Apply**.

8 Open Command Prompt as an administrator and run `gpupdate /force`. Press **Enter** on your keyboard.

Installing Mission Control 3.4.1.0

To upgrade your Security Center system to a collaborative decision management system, you must install Genetec Mission Control™ on all Security Center servers hosting the Incident Manager role.

Before you begin

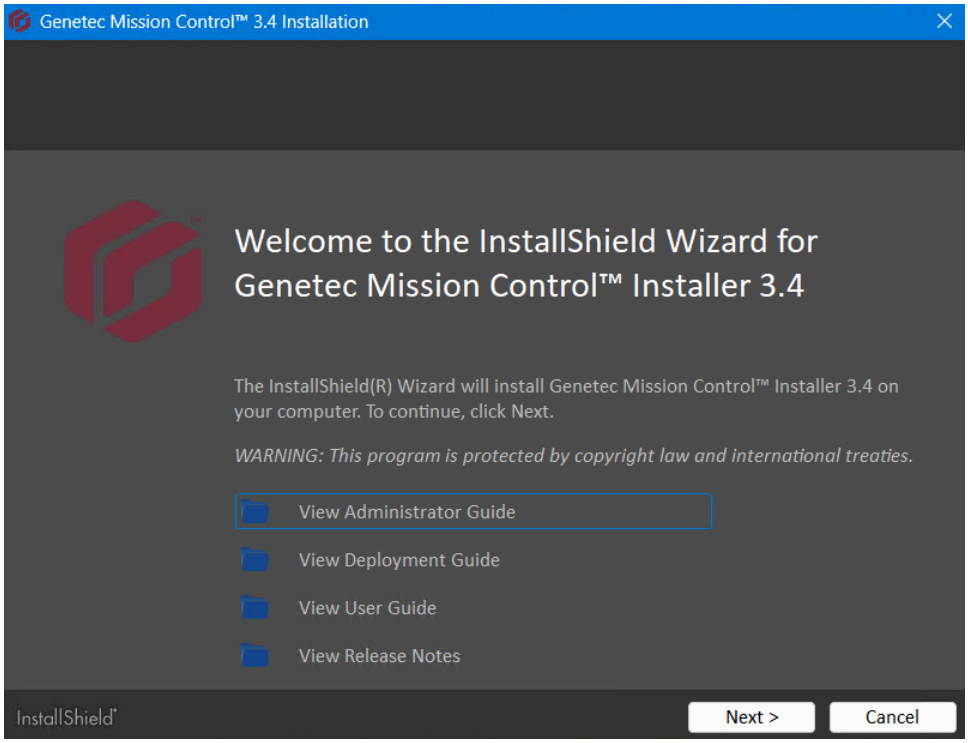
- Download Mission Control.

Procedure

1 Launch the Mission Control installer.

2 On the *Choose Setup Language* page, select English or French and click **Next**.

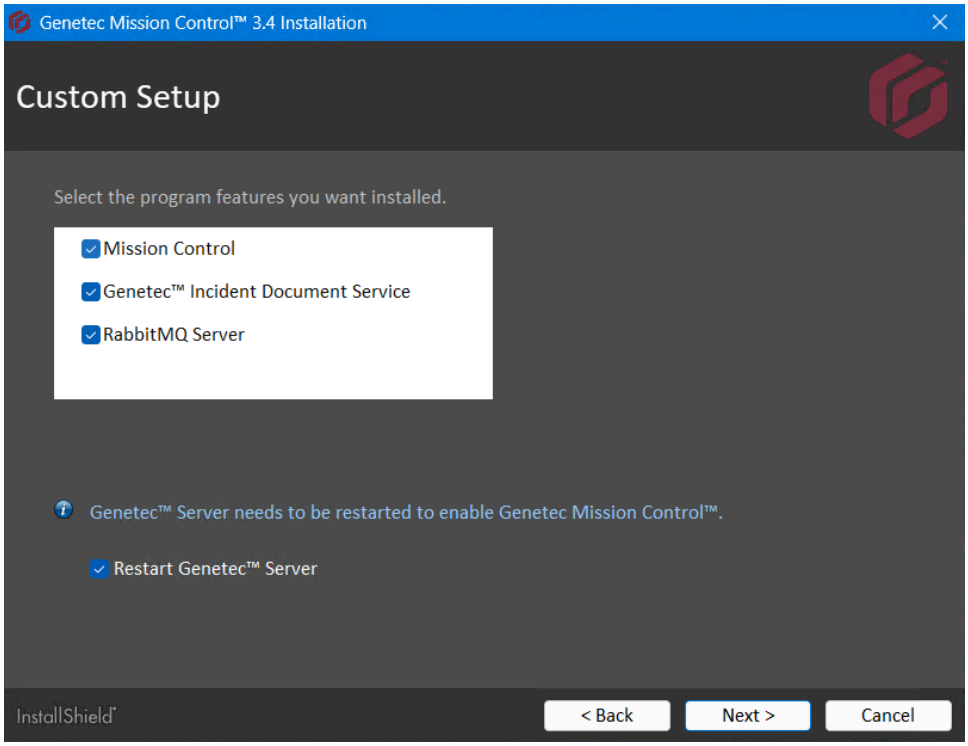
3 On the *Welcome to the InstallShield Wizard* page, click the links to read Mission Control documentation, and then click **Next**.



4 On the *License Agreement* page, read the terms in the *Genetec™ Software License Agreement*, select **I accept the terms in the license agreement**, and then click **Next**.

5 In the **Custom Setup** page, select the applications that you want to install.

A custom setup is useful when you want to install one of the applications on another server for high availability.



- **Mission Control:** Installed on all servers that host the Incident Manager role on all client workstations.
- **Genetec™ Incident Document Service:** Installed on all servers that host the Incident Manager role.

  It's required to enable adding documents to an incident in Security Desk.
- **RabbitMQ Server:** Can be installed on any server, not necessarily a Security Center server.

  If RabbitMQ is already installed on this machine for another product, clear this option.

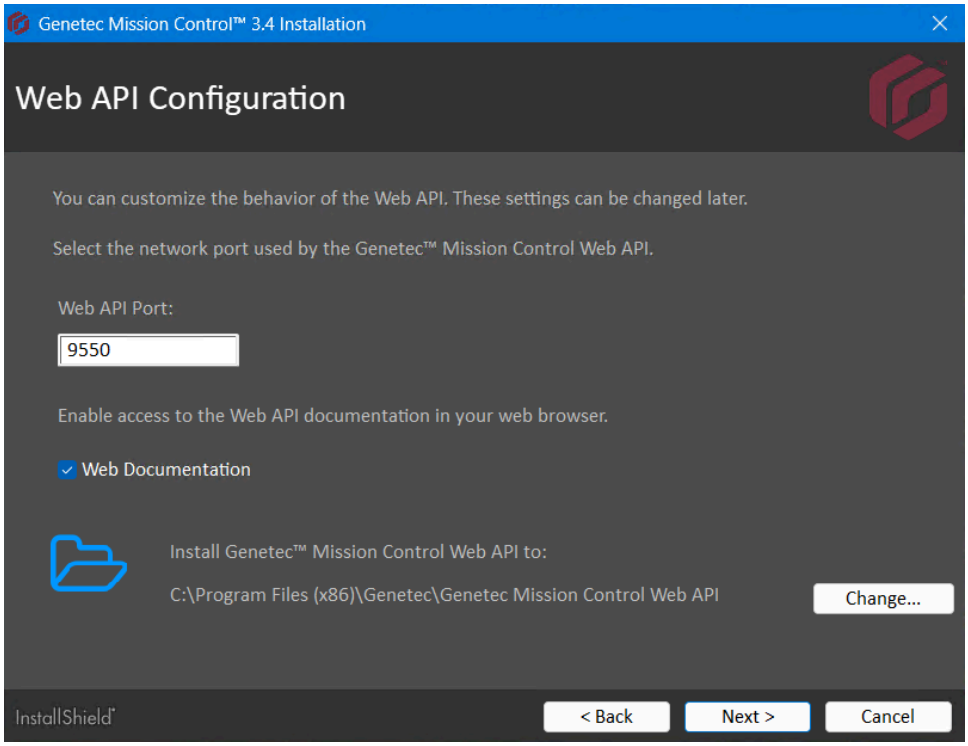
  If you choose to install RabbitMQ, the latest version is installed.

  TIP: Don't change the existing RabbitMQ hostname.

NOTE: The option to **Restart Genetec™ Server** after installation of selected applications is selected by default. If you clear this option, you can manually restart the Genetec™ Server later.

6 Click **Next**.

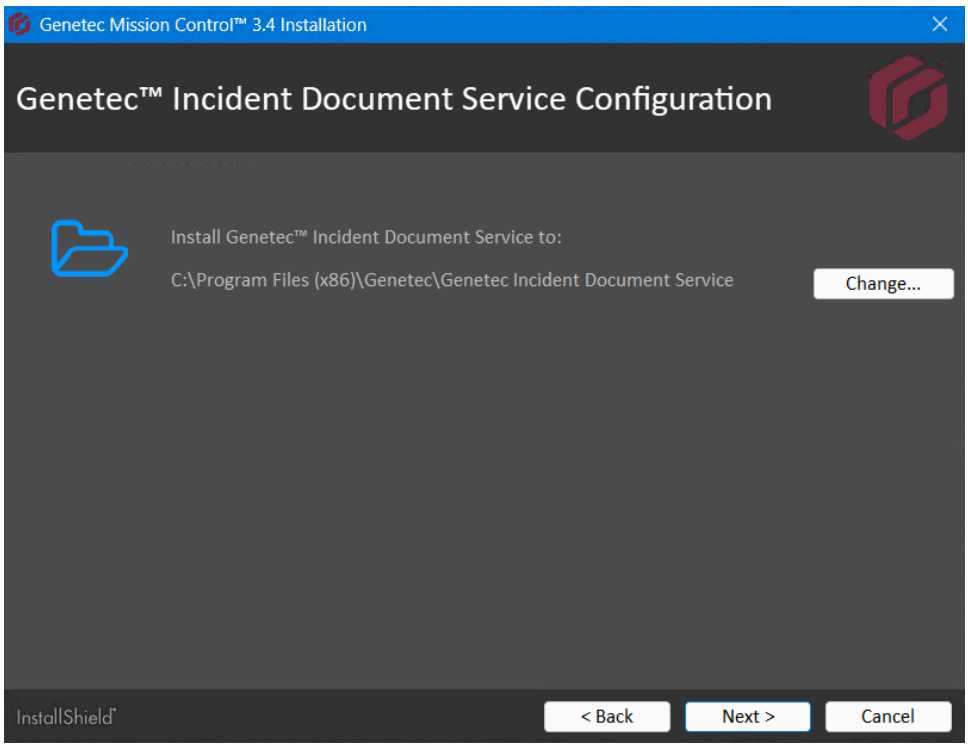
7 In the *Web API Configuration* page, select the **Web Documentation** checkbox to access Web API documentation.



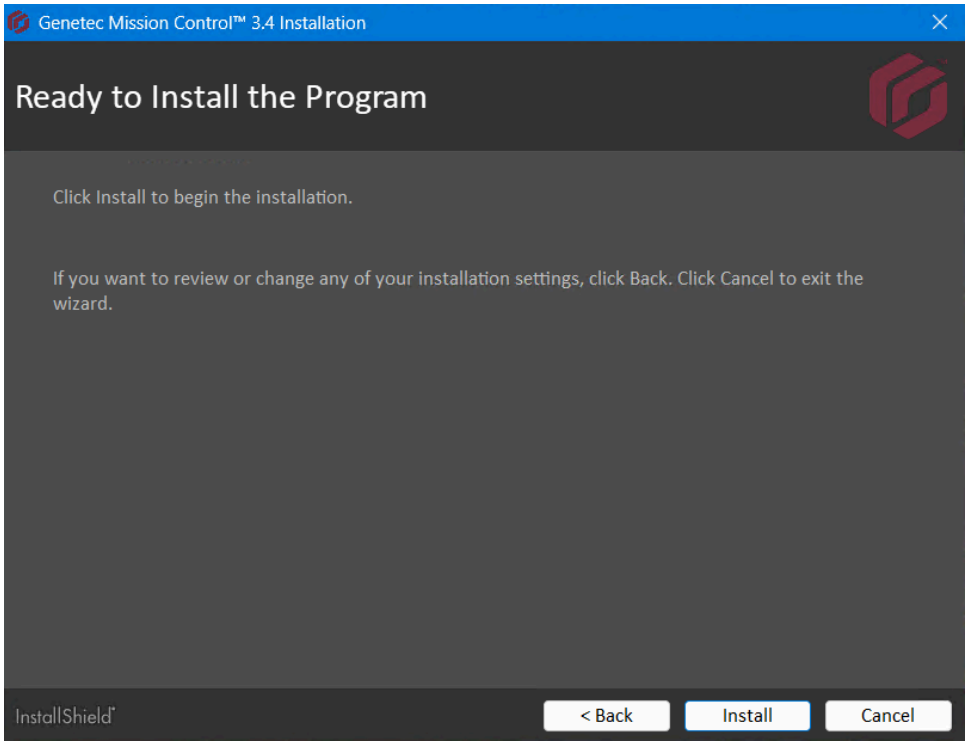
NOTE: (Optional) You can change the default **Web API port** and install location for the Mission Control Web API. The default port is 9550.

BEST PRACTICE: Disable web documentation in a production environment.

8 In the *Genetec™ Incident Document Service Configuration* page you can configure the default location to install Genetec Incident Document Service.

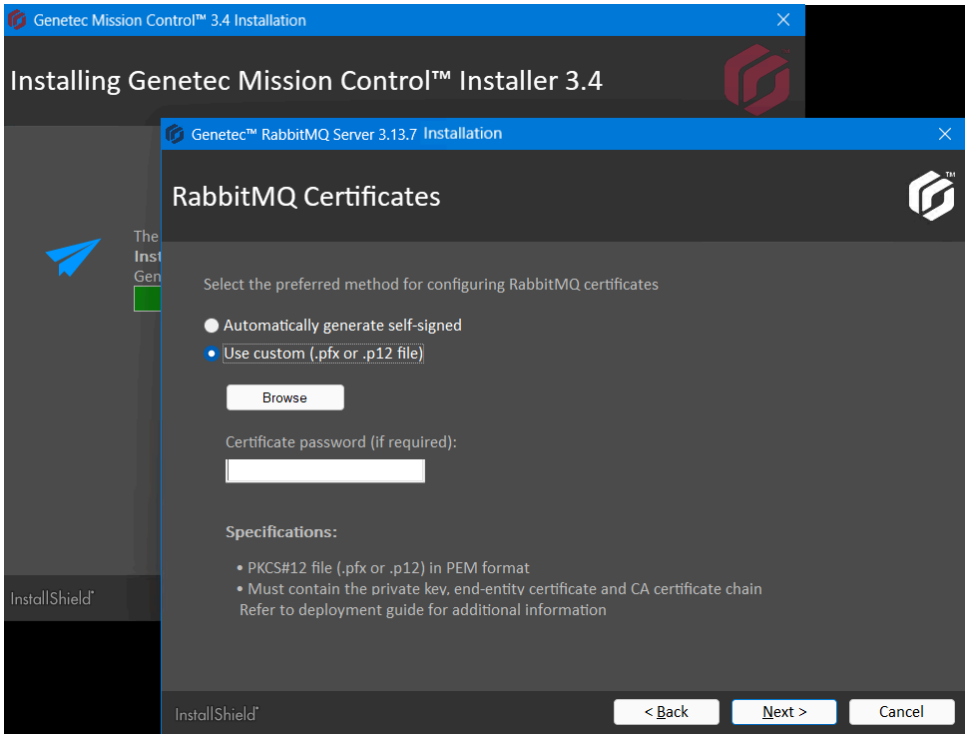


9 Click **Next** and you're ready to install the program.



10 Click **Install**.
A separate installer is launched to configure RabbitMQ.

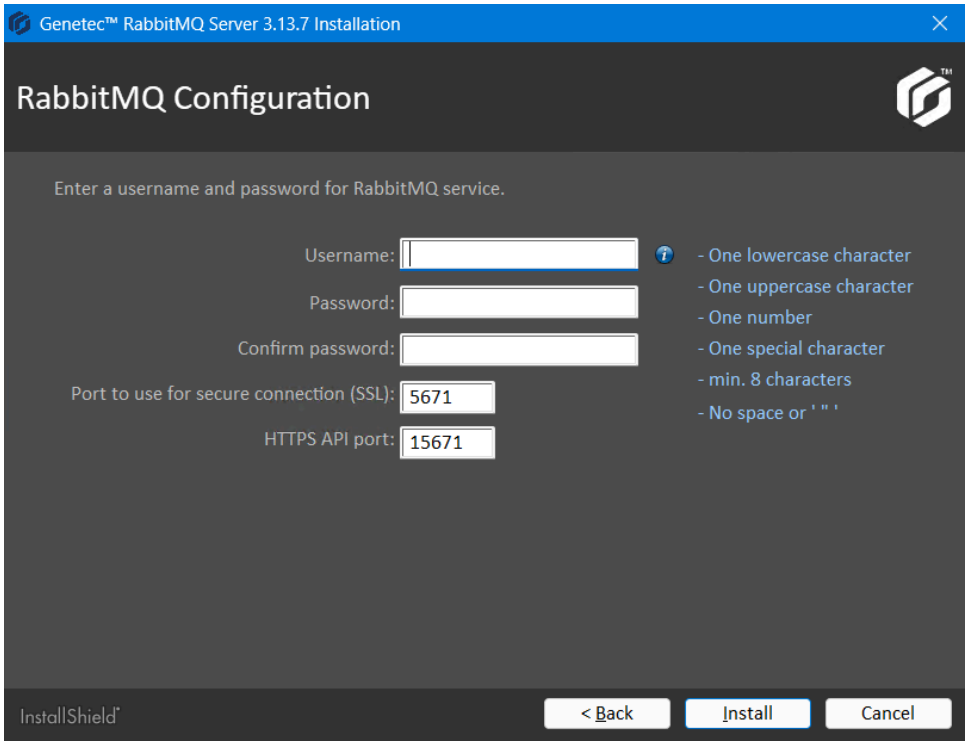
11 In the *RabbitMQ Certificates* page, select the method for configuring certificates.



- **Self-signed:** The certificate is generated automatically.
- **Custom:** Upload a custom *.pfx* or *.p12* file in PEM format. It must contain a private key, end-entity certificate and a CA certificate chain.

12 Click **Next**.

13 Enter the credentials for the RabbitMQ server on the *RabbitMQ Configuration* page:



- **Username and Password:** Credentials used by the Incident Manager to connect to the RabbitMQ server.
- **Port to use for secure connection (SSL):** Communication port for connections between Security Center applications and the RabbitMQ server. The default port is 5671 but you can modify it as required.
- **HTTPS API port:** Communication port used for secure HTTPS connections. The default port is 15671 but you can modify it as required.

BEST PRACTICE: Ensure optimal performance of the RabbitMQ service and avoid potential issues by configuring your anti-virus software to exclude the RabbitMQ database folder during scans.

Folder location: *C:\ProgramData\Genetec\RabbitMQ\db*.

NOTE:

- The Incident Manager role can only connect to RabbitMQ using the SSL port. Using the non-SSL port results in a failed connection. For instructions on modifying the port assignment, see Modifying RabbitMQ ports on page 62.
- It is recommended to install *Handle.exe* on all systems using RabbitMQ to avoid connection and latency issues. It is not included in the Mission Control installer due to Microsoft software license terms. For information about installing *Handle.exe*, see Install Handle.exe for RabbitMQ.

14 Click **Install**.

15 If you didn't select the option to restart the Genetec Server service as part of the installation, restart it after the installation is complete.

16 Click **Close**.

The following are added to Security Center:

- New privileges for Mission Control
- New privileges for Mission Control Web API

- New tasks that are enabled by the license:
  - *Incident configuration*
  - *Incident monitoring*
  - *Incident report*
  - *Profile report*

After you finish

- Create the Incident Manager role.
- Create the Incident Document Service role.
- Assign Mission Control privileges to your users.

  BEST PRACTICE: If you are upgrading from version 3.2.0.0 or earlier, verify that all your documents have been migrated to the SQL Server before you uninstall MongoDB from your system.

Modifying RabbitMQ ports

By default, RabbitMQ uses port 5671 for SSL communication. If you want to change the default port numbers used by Genetec Mission Control™, you must either change it during the installation process or change the port in the *RabbitMQ.conf* file.

Procedure

1 Update the *RabbitMQ.conf* file.
   a) On the machine where Mission Control is installed, navigate to *Computer\Local Disk (C:)\ProgramData\Genetec\RabbitMQ\conf*.
   b) Open the *RabbitMQ.conf* file in a text editor.
   c) To change the **SSL** port from the default (5671 is selected by default during installation), modify the line:

```
listeners.ssl.default = 5671
```

   d) To change the HTTPS API port from the default (15671), modify the line:

```
management.ssl.port       = 15671
```

   e) Save and close the *RabbitMQ.conf* file.

2 In the **Services** tab of the Windows Task Manager, restart the RabbitMQ service.

3 Update the ports in the Incident Manager role to match the port information in the *RabbitMQ.conf* file.
   a) From the Config Tool homepage, open the *System* task, and click the **Roles** view.
   b) Select the Incident Manager role, and click the **RabbitMQ** tab.
   c) Select the server you want to modify and click **Edit the item** ( ).
   d) Enter the new port number and click **Apply**.

      NOTE: You can modify either or both of the port information depending.
   e) If you have modified the ports on more than one RabbitMQ server, and if those servers are listed, then you must also edit the port for those servers.

Certificates for RabbitMQ

The Genetec Mission Control™ installer comes with self-signed certificates for use with RabbitMQ.

RabbitMQ configuration file

By default, RabbitMQ uses the self-signed certificates packaged with the Mission Control installer. The certificate details can be found in *C:\ProgramData\Genetec\RabbitMQ\config\RabbitMQ.conf*:

```
## HTTPS listener settings.
## See https://rabbitmq.com/management.html and https://rabbitmq.com/ssl.html for
 details.
##
 management.ssl.port       = 15671
 management.ssl.cacertfile = C:/Program Files (x86)/Genetec/RabbitMQ/CACert.pem
 management.ssl.certfile   = C:/Program Files (x86)/Genetec/RabbitMQ/Cert.pem
 management.ssl.keyfile    = C:/Program Files (x86)/Genetec/RabbitMQ/Key.pem
 management.ssl.honor_cipher_order   = true
 management.ssl.honor_ecc_order      = true
 management.ssl.client_renegotiation = false
 management.ssl.secure_renegotiate   = true
## Supported TLS versions
 management.ssl.versions.1 = tlsv1.2
## Cipher suites the server is allowed to use
 management.ssl.ciphers.1  = ECDHE-ECDSA-AES256-GCM-SHA384
 management.ssl.ciphers.2  = ECDHE-RSA-AES256-GCM-SHA384
 management.ssl.ciphers.3  = ECDH-ECDSA-AES256-GCM-SHA384
 management.ssl.ciphers.4  = ECDH-RSA-AES256-GCM-SHA384
 management.ssl.ciphers.5  = DHE-RSA-AES256-GCM-SHA384
 management.ssl.ciphers.6  = DHE-DSS-AES256-GCM-SHA384
 management.ssl.ciphers.7  = ECDHE-ECDSA-AES128-GCM-SHA256
 management.ssl.ciphers.8  = ECDHE-RSA-AES128-GCM-SHA256
 management.ssl.ciphers.9  = ECDH-ECDSA-AES128-GCM-SHA256
 management.ssl.ciphers.10 = ECDH-RSA-AES128-GCM-SHA256
 management.ssl.ciphers.11 = DHE-RSA-AES128-GCM-SHA256
 management.ssl.ciphers.12 = DHE-DSS-AES128-GCM-SHA256
```

Creating custom certificate requests

If you choose to integrate third-party certificates into RabbitMQ, you must make a certificate signing request (CSR).

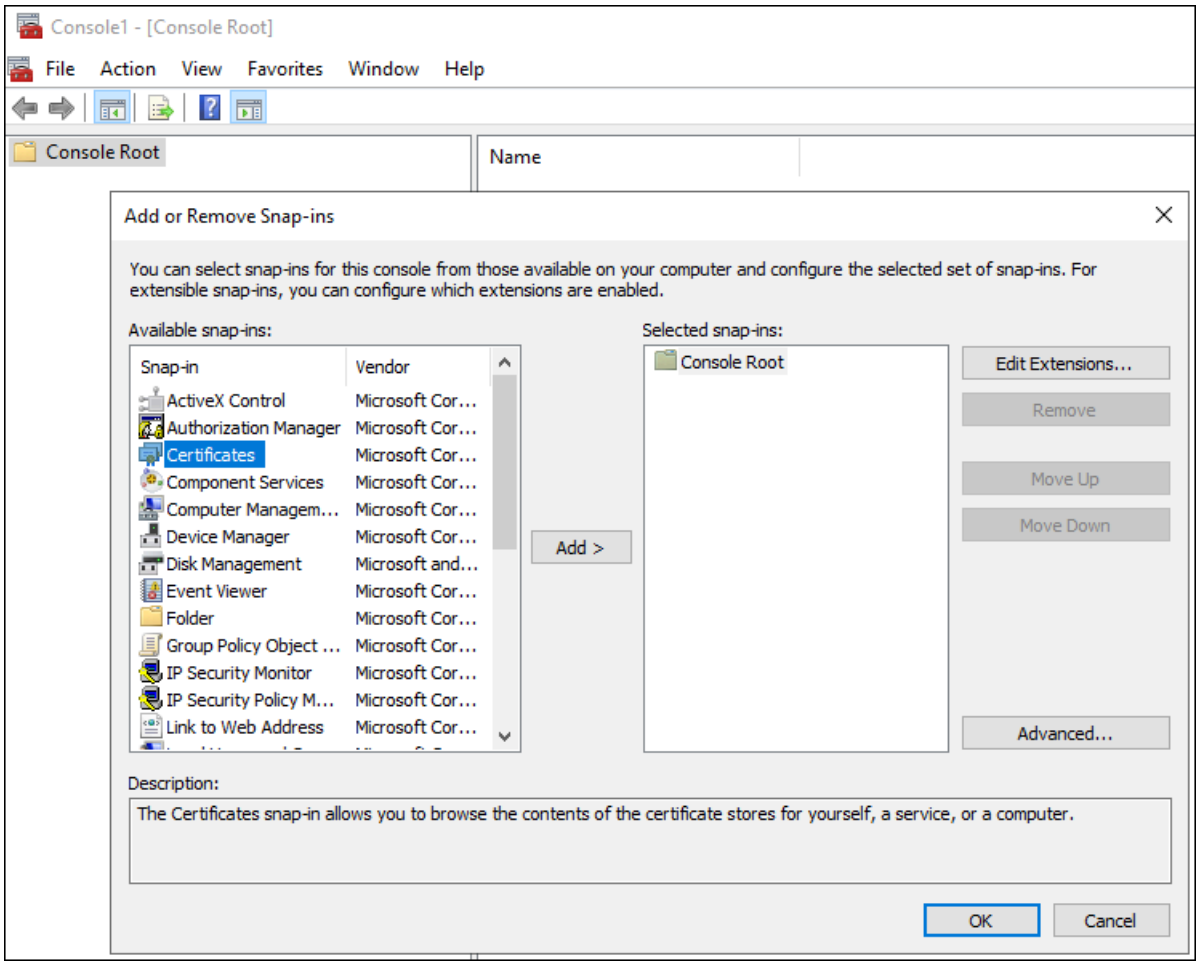
Procedure

To request a certificate signing request (CSR) file:

1 Open Windows Console:
   a) In the Windows search box, type Run.
   b) In the *Run* dialog box, type `mmc.exe`.
      The *Console* window opens.

2 Open the *Add/Remove Snap-in* task by selecting **File** > **Add/Remove Snap-in**.

3 Add certificates from the *Available snap-ins* list to the *Selected snap-ins* list:
   a) Select the required certificates.
   b) Click **Add**.
   c) In the *Certificates snap-in* dialog box, select **Computer account** and click **Next**.
   d) Select **Local computer** and click **Finish**.



   You are directed back to the *Add or Remove Snap-ins* window.

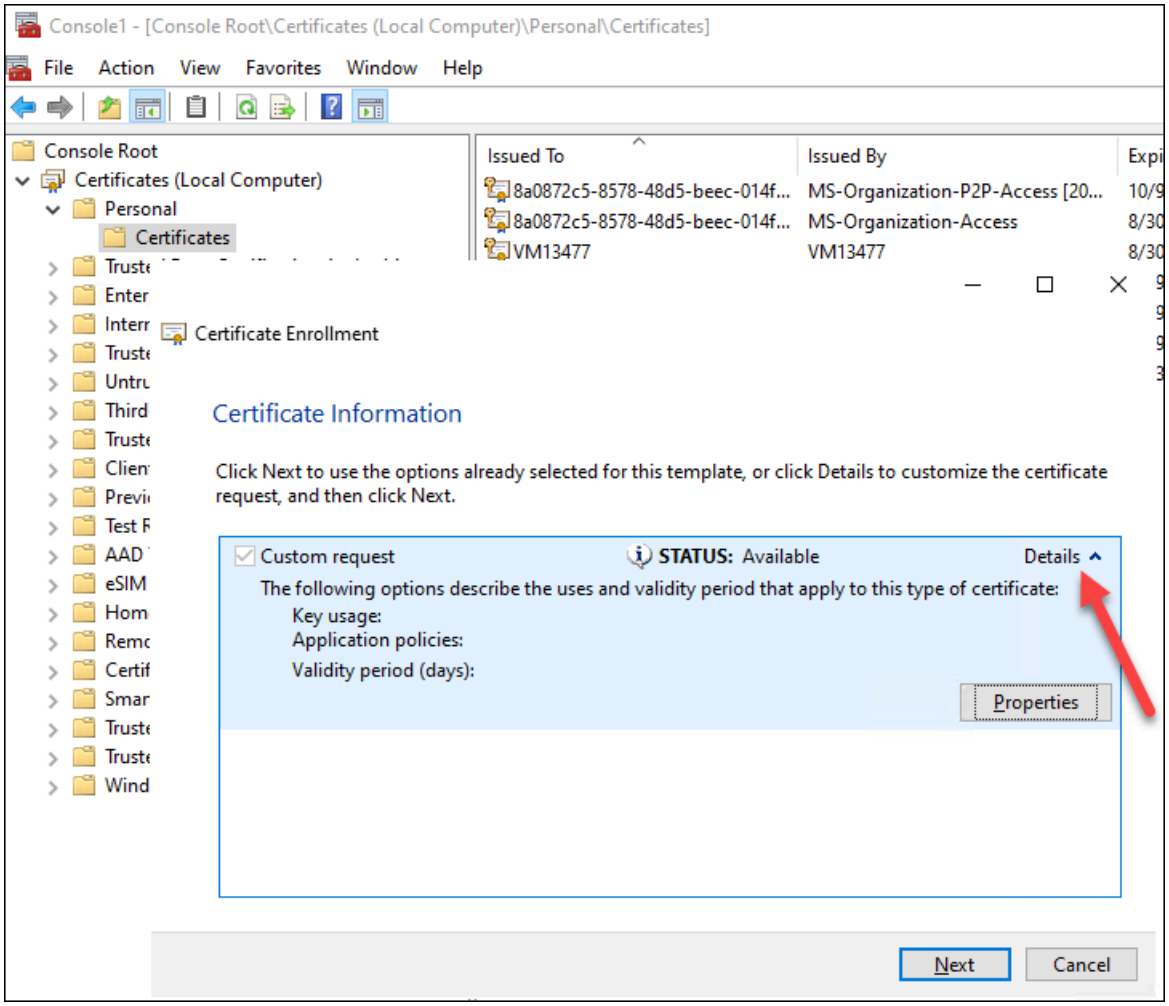
4 Click **OK**.
   Certificates (Local Computer) is added under **Console Root** in the *Console* window.

To enroll the certificates:

1 In the *Console* window, expand the newly added **Certificates** entry.
2 Navigate to **Certificates (Local Computer)** > **Personal** > **Certificates**.
3 Right-click the **Certificates** folder and select **All Tasks** > **Advanced Operations** > **Create Custom Request**.
   The *Certificate Enrollment* window opens.
4 Click **Next**.
5 Select **Proceed without enrollment policy** and click **Next**.
6 Select the following parameters and then click **Next**:
   - From the **Template** list, select **(No template) Legacy key**.
   - Select **PKCS#10** for **Request format**.

   The *Certificate Enrollment* dialog box shows the certificate information.

7 Expand the **Details** and then click **Properties**.



8 In the *Certificate properties* dialog box, click the *Subject* tab and enter the following details:

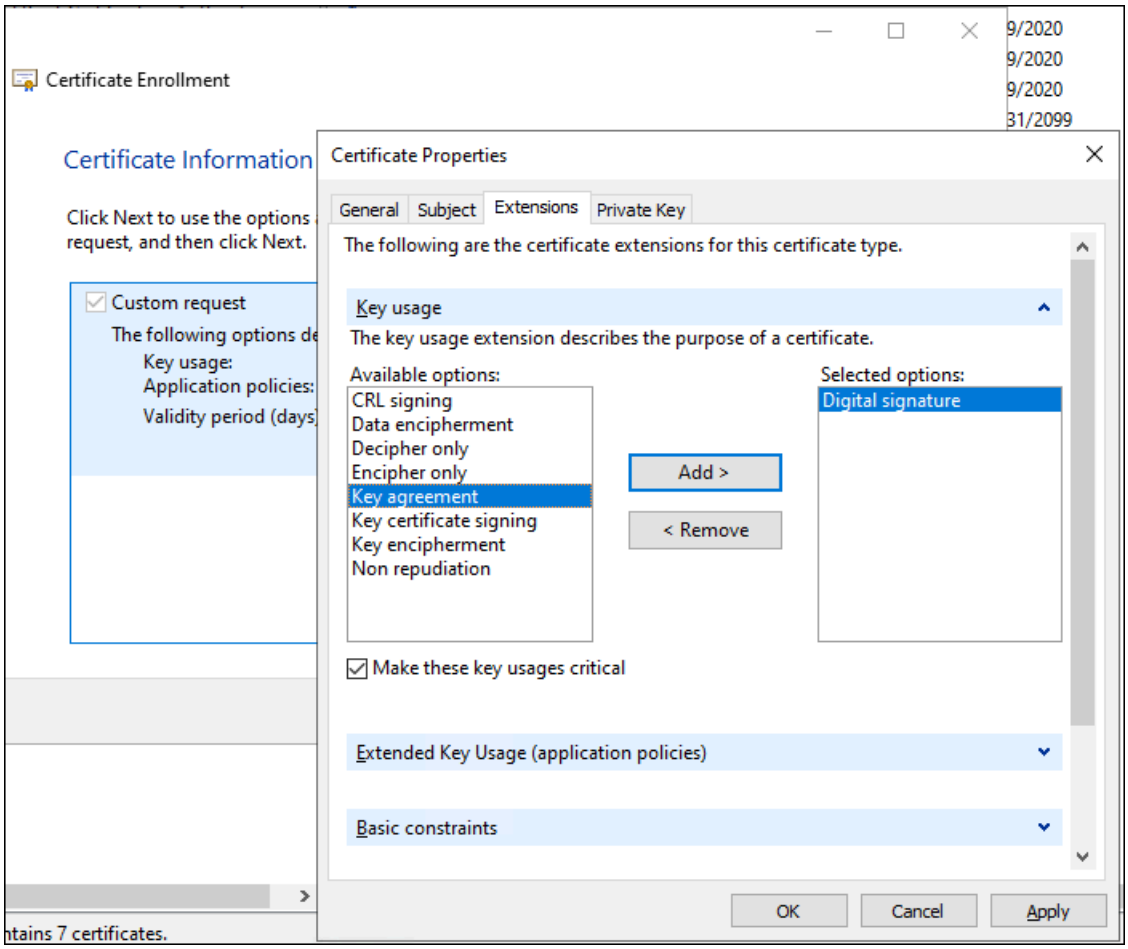
- From the **Type** list, select **Common name**.
- In the **Value** field, enter `<YourServerName>.<YourCompanyDomain.com>`.

IMPORTANT: The common name you enter must match the fully qualified domain name of the server. For example, if the hostname of your server is *Server1* and your domain is *MyCompany.com*, then the fully qualified domain name of your server would be *Server1.MyCompany.com*.

9 Click the *Extensions* tab and perform the following actions:

a) Expand **Key usage** and add the following options to the *Selected options* list:

- **Digital Signature**
- **Key agreement**



b) Expand **Extend key usage (application policies)** and add the following to the *Selected options* list:

- **Server Authentication**
- **Client Authentication**

10 Click the **Private Key** tab and set the following properties:

a) Expand **Key type** and select **Exchange**.

b) Expand **Key options**, set **Key size** to **2048**, and select **Make private key exportable**.

c) Expand **Cryptographic Service Provider** and select **Microsoft RSA SChannel Cryptographic Provider (Encryption)**.

This is the last option on the list.

11 Click **Apply** and then click **OK**.

12 In the *Certificate Enrollment* dialog box, click **Next**, and save your certificate request:

a) Select a name and destination for your file.

b) Choose **Base 64** for the **File format**.

13 Click **Finish**.

Your request file is saved in the *.csr* format.

After you finish

- Send your request file to your IT department or external certificate authority for processing.
- After the certificate has been generated, you can import it and apply it to your server.

Integrating custom certificates into RabbitMQ

When using third-party certificates, you need a private and a public key to enable integration of the certificates with RabbitMQ.

Procedure

To install the certificates on the local machine:

1 Open the Windows Console:
   a) In the Windows search box, type `Run`.
   b) In the *Run* dialog box, type `mmc.exe` and click **OK**.
      The *Console* window opens.

2 In the left pane of the *Console* window, expand **Personal** and click **Certificates**.

3 In the right pane of the window, right-click the certificate you created and click **All Tasks Export**.
   The *Certificate Export Wizard* opens.

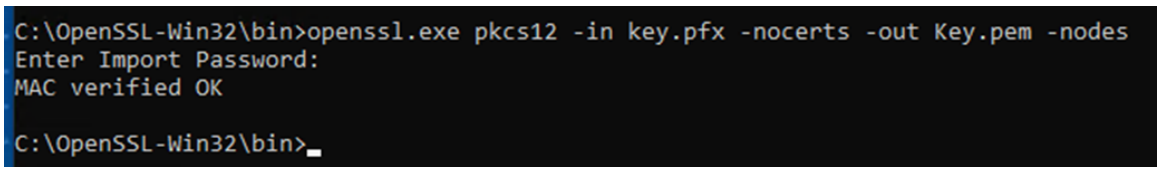
4 Select the following options:
   a) On the *Export Private Key* page, select **Yes, export the private key** and click **Next**.
   b) On the *Export File Format* page, select **Personal Information Exchange**, clear **Enable certificate privacy** and click **Next**.
   c) On the *Security* page, select **Password**, type your password choice, and click **Next**. Note your password as you need it to import the certificate.
   d) Name your certificate export file (.pfx) and click **Finish**.

To convert the PFX file to a PEM file:

1 Check if OpenSSL is installed in *C:\Program Files (x86)\OpenSSL-Win32* or install it from Downloads | OpenSSL Library.

2 Open the command prompt as an administrator:
   a) In the Windows search box, type `cmd`.
   b) Right-click **Command Prompt** and select **Run as administrator**.

3 Generate the *key.pem* file:
   a) Change the directory using the following command: `c:\OpenSSL-Win32\bin`
      NOTE: Copy your PFX file to this location.
   b) Run `Openssl.exe pkcs12 -in yourpfxfile.pfx -nocerts -out key.pem -nodes`
   c) When prompted for a password, enter the password you set up when exporting the certificates from *mmc.exe*.
   The *key.pem* file is generated in *C:\OpenSSL-Win32\bin*.



4 Move the *Key.pem* file to *C:\Program Files (x86)\Genetec\RabbitMQ*.

To export the .Cert files

1 Open the Windows Console:
   a) In the Windows search box, type `Run`.
   b) In the *Run* dialog box, type `mmc.exe` and click **OK**.
      The *Console* window opens.
2 In the left pane of the *Console* window, expand **Personal** and click **Certificates**.
3 In the right pane of the window, right-click the certificate you created and click **All Tasks Export**.
   The *Certificate Export Wizard* opens.
4 Select the following options:
   a) On the *Export Private Key* page, select **No, do not export the private key** and click **Next**.
   b) On the *Export File Format* page, select **DER encoded binary X.509 (.CER)** and click **Next**.
      The name of this export file is CACert.cer.
   c) On the *Export Private Key* page, select **No, do not export the private key** and click **Next**.
   d) On the *Export File Format* page, select **Base-64 encoded X.509 (.CER)** and click **Next**.
      The name of this export file is Cert.cer.

To complete the certificate integration

1 Back up your RabbitMQ self-signed certificate files in *C:\Program Files (x86)\Genetec\RabbitMQ*.
2 Replace the RabbitMQ self-signed certificates:
   a) Go to *C:\Program Files (x86)\Genetec\RabbitMQ*.
   b) Replace *Cert.cer* with *Cert.pem*.
      NOTE: You can change the extension by renaming it.
   c) Replace *CACert.cer* with *CACert.pem*.
3 Open the **RabbitMQ Command Prompt (sbin dir)** application.
4 Reinstall the RabbitMQ service by running the following commands:
   a. `rabbitmq-service.bat remove`
   b. `rabbitmq-service.bat install`
   c. `rabbitmq-service.bat start`
5 Reboot your machine for the changes to take effect.

Installing the Mission Control patch release

The patch release must be installed on all Security Center servers hosting the Incident Manager role.

Before you begin

- Ensure that you have the latest minor version of Genetec Mission Control™ installed on your system.
  TIP: For example, if you are upgrading to Mission Control 3.2.1.1, ensure you have Mission Control 3.2.1.0 before installing the patch release.
- Ensure that Security Center and RabbitMQ are compatible with the minor version of Mission Control. See the Product compatibility for Mission Control article for the latest information.

Procedure

1 Log on to the Genetec Portal and download the patch release:
   a) Click > **Technical Assistance** > **Product Download** .
   b) From the **Download Finder** list, select **All Products**.
   c) Download the Mission Control <*PATCH RELEASE VERSION*> package.

2 Close all applications (Config Tool, Security Desk, and so on).

3 If you're updating your main server, stop the Genetec™ Server service:
   a) Open your Windows Control Panel.
   b) Click **Administrative Tools** > **Services**.
   c) Select the **Genetec™ Server** service, and then click **Stop**.

4 Right-click the *.exe* file and select **Run as administrator**.

5 After *InstallShield* finishes analyzing your computer, click **Update** to start the installation.

6 When the installation is complete, click **Finish** and restart your system to complete the installation.

7 If you stopped the Genetec Server service, restart it from the *Services* window by selecting **Genetec™ Server** and clicking **Start**.

Installing Mission Control in silent mode

You can use the Genetec Mission Control™ installer program, *MCInstaller.exe*, for silent mode installation.

Before you begin

- The Security Center silent install command must be executed before the Mission Control silent install command in your batch files.
- If installing a patch release, ensure that you have the latest minor version of Genetec Mission Control™ installed on your system.

  TIP: For example, if you are upgrading to Mission Control 3.2.1.1, ensure you have Mission Control 3.2.1.0 before installing the patch release.
- Ensure that Security Center and RabbitMQ are compatible with the minor version of Mission Control. See the Product compatibility for Mission Control article for the latest information.

Procedure

To install Mission Control in silent mode:

1 Run the following command line from the Mission Control installation package:

```
"MC Packages\MCInstaller.exe" /silent /ISFeatureInstall=<Features> [<LogOptions>]
```

This generic command allows you to install Mission Control and all its components together or individually.

To install the Mission Control patch release in silent mode:

1 Open a command prompt with elevated privileges as an administrator.

2 If you are updating your main server, stop the Genetec™ Server service by entering: `net stop GenetecServer`

3 Start the installer by entering: `"<NAME_OF_PATCH_RELEASE>.exe" /s /v"/qn"`

The command can also be run by providing the complete file path:

```
"C:\Genetec\<NAME_OF_PATCH_RELEASE>.exe" /s /v"/qn"
```

NOTE: The system restarts automatically to complete the installation.

4 If you stopped the Genetec™ Server service, restart it by entering: `net start GenetecServer`

Related Topics

Silent install command syntax for Mission Control on page 70

Silent install command syntax for Mission Control

When performing a silent installation, specific program options are required to run the Genetec Mission Control™ Installer.

The syntax for running the Mission Control installer in silent mode is:

```
"MC Packages\MCInstaller.exe" <installer_options>
```

The following table lists the installer options:

Installer option	Description
`/ISFeatureInstall`	Specifies the features to be installed. The possible values must be separated by a comma without spaces: • `Server` (Genetec Mission Control™ feature) • `DocServer` (Genetec™ Incident Document Service feature) • `RabbitMQ` (RabbitMQ feature) Must be used with: • `RABBITUSER=<username>` • `RABBITPWD=<password>` • `CONFIRMPWD=<password>` These three options can be put in any order on the command line. **EXAMPLES**: • Install Mission Control, Mission Control Web API, Incident Document Service and RabbitMQ Server: `/ISFeatureInstall=Server,DocServer,RabbitMQ ISInstallDir_MissionControlWebAPI="C:\Program Files (x86)\Genetec\Genetec Mission Control Web API" RESTART_GENETEC_SERVER=1 GENETEC_PORT=9550 ENABLE_WEBAPI_DOCUMENTATION=1 ISInstallDir_DocumentService="C:\Program Files (x86)\Genetec\Genetec Document Service RABBITUSER=localadmin RABBITPWD=Password_1! CONFIRMPWD=Password_1! SSLPORT=5671 MANPLUGPORT=15671"` • Install Mission Control only: `/ISFeatureInstall=Server ISInstallDir_MissionControlWebAPI="C:\Program Files (x86)\Genetec\Genetec Mission Control Web API" RESTART_GENETEC_SERVER=1 GENETEC_PORT=9550 ENABLE_WEBAPI_DOCUMENTATION=1"` • Install Incident Document Service only: `/ISFeatureInstall=DocServer ISInstallDir_DocumentService="C:\Program Files (x86)\Genetec\Genetec Document Service"` • Install RabbitMQ Server only: `/ISFeatureInstall=RabbitMQ RABBITUSER=localadmin RABBITPWD=Password_1! CONFIRMPWD=Password_1! SSLPORT=5671 MANPLUGPORT=15671` **EXAMPLE OF INCORRECT SYNTAX**: • `/ISFeatureInstall=RabbitMQ RABBITUSER=Admin RABBITPWD=sEcReT CONFIRMPWD=sEcReT, Server, WebAPI` `RabbitMQ`, and `Server` must immediately follow `/ISFeatureInstall`.
`/silent`	Sets the Mission Control installer to run in silent mode with no user interaction.

Installer option	Description
`/debuglog<FilePath>`	Enables the creation of the debug log file and specifies the file path. **NOTE:** The file path specified in `<FilePath>` must exist. The installer will not create it. **EXAMPLE**: `/debuglog"C:\DebugLog.log"`
`/log<FolderPath>`	Enables the creation of log files and specifies the file path. **NOTE:** The file path specified in `<FilePath>` must exist. The installer will not create it. **EXAMPLE**: `/log"C:\MyLogFiles\"`
`/remove`	Removes the installed features.

Related Topics

Installing Mission Control in silent mode on page 69

Creating the Incident Manager role

Before you can configure and use Genetec Mission Control™, you must create the Incident Manager role in Config Tool.

Before you begin

Ensure you have installed Mission Control.

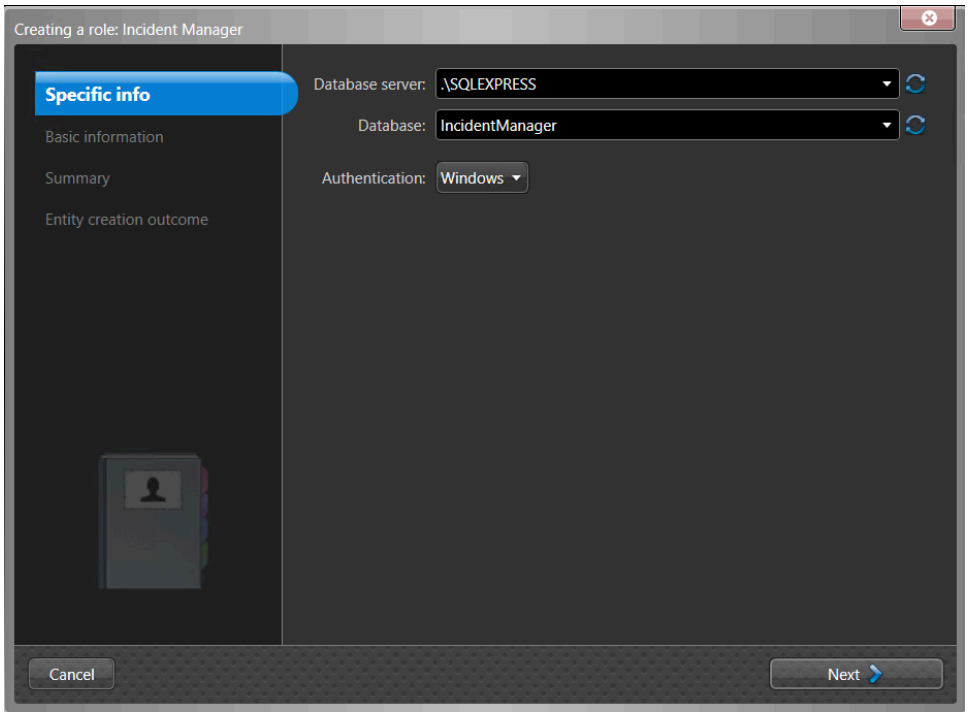
What you should know

Procedure

1 From the Config Tool homepage, open the *System* task and click the **Roles** view.

2 Click **Add an entity** ( ) and then click **Incident Manager** ( ).

The *Creating a role: Incident Manager* window opens.



NOTE: The Incident Manager role is hosted by default on your main server and there can be only be only one instance of this role on your system.

3 On the *Specific info* page, do the following:

a) In the **Database server** field, select or type the name of the database server.

b) In the **Database** field, select or type the name of the database (for example, **IncidentManager**).

c) In the **Authentication** field, choose **Windows** or **SQL Server** as your authentication method.

d) Click **Next**.

4 On the *Basic information* page, do the following:

a) Type the **Entity name** (**Incident Manager**).

b) Type an **Entity description** for the role.

c) Click **Next**.

5 On the *Summary* page, do the following:

a) Verify the information you entered.

b) If everything is correct, click **Create**, or click **Back** to modify your settings.

When the role is created, you see the following message in the **Entity creation outcome** page:

```
The operation was successful.
```

NOTE: Incident Manager role configuration can also be modified after the role has been created.

After you finish

- Create the Incident Document Service role.
- Assign privileges to your users.
- Configure the Incident Manager role.
- Validate the RabbitMQ configuration.

Creating the Incident Document Service role

To enable adding documents to an incident in Security Desk, you must create the Incident Document Service role in Config Tool.

Before you begin

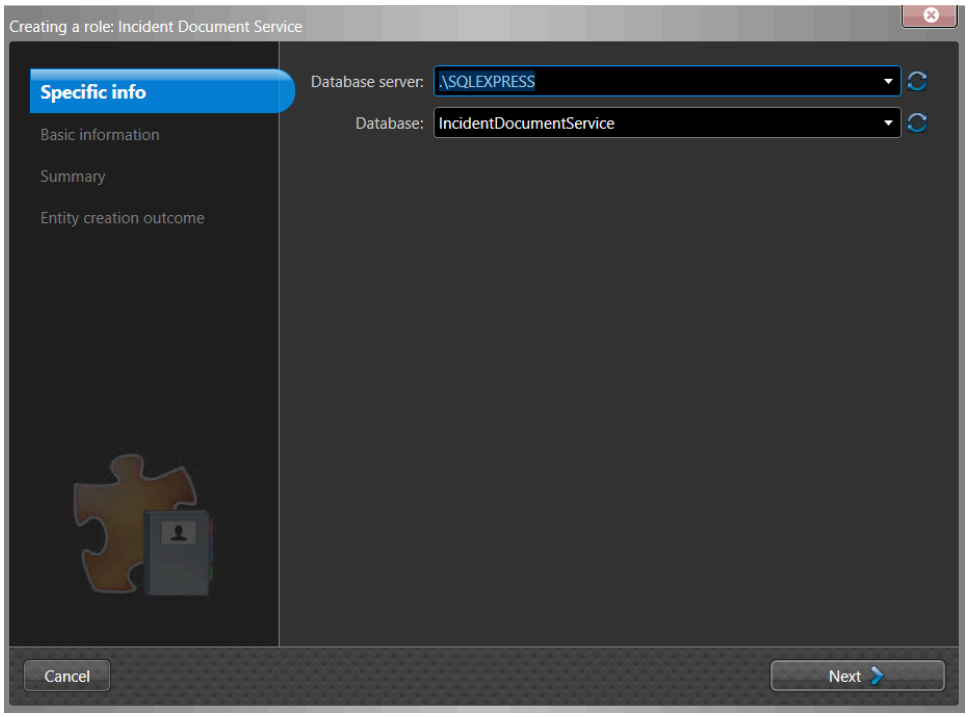
Ensure you have installed Mission Control.

Procedure

1 From the Config Tool homepage, open the *System* task and click the **Roles** view.

2 Click **Add an entity** ( ) and then click **Incident Document Service** ( ).

The *Creating a role: Incident Document Service* window opens.



3 On the *Specific info* page, do the following:

a) In the **Database server** field, select or type the name of the database server.

b) In the **Database** field, select or type the name of the database (for example, **IncidentDocumentService**).

c) Click **Next**.

4 On the *Basic information* page, do the following:

a) Type the **Entity name** (**Incident Document Service**).

b) Type an **Entity description** for the role.

c) Click **Next**.

5 On the *Summary* page, do the following:

a) Verify the information you entered.

b) If everything is correct, click **Create**, or click **Back** to modify your settings.

When the role is created, you see the following message in the **Entity creation outcome** page:

```
The operation was successful.
```

NOTE: Incident Document Service role configuration can also be modified after the role has been created.

Mission Control privileges

For administrators to configure incidents in Config Tool, and for operators to use the *Incident monitoring* and *Incident report* tasks in Security Desk, the correct user privileges must be granted to their user accounts.

Configuring user privileges for Genetec Mission Control™

Default privileges are assigned to some user groups, such as administrators, operators, and supervisors. Individual users in these user groups inherit their privileges from the parent user group, unless specified otherwise. The operator and supervisor privilege template typically grants access to Security Desk.

In the Config Tool *User management* task, you can configure or modify the user or user group privileges on the *Privileges* page of the user or user group.

To learn more about the privilege hierarchy, privilege inheritance, and assigning privileges, see the Security Center Administrator Guide and the Security Center Hardening Guide.

NOTE: For a list of all available Security Center privileges, see the Security Center privileges spreadsheet. You can sort and filter this list as needed.

License options

A license is consumed when a user with Mission Control privileges logs on to Security Desk and does any of the following:

- Open the *Incident monitoring* task.
- Open the *Incident report* task.
- Open an incident from the notification tray.

Administrative privileges

Administrative privileges grant access to Mission Control incident management activities using Config Tool.

By default, administrators have all privileges. If you create a user account from one of the other privilege templates and you want this user to add and configure incidents, the user account requires the following incident management administrative privileges for Config Tool in Mission Control:

Privilege	Subcategory of privileges	Includes privileges to	Actions that can be performed
Incident management (Genetec Mission Control™)	View incident and incident category properties	Modify incident and incident category properties	• Add incidents and incident categories • Delete incidents and incident categories • Modify incident types and categories operational status
	View document properties	Modify document properties	

Privilege	Subcategory of privileges	Includes privileges to	Actions that can be performed
	View profile properties	Modify profile properties	• Add profiles • Deactivate profiles • Delete profiles
	View incident audit trail report		

Action privileges

Action privileges control the actions that can be performed on Mission Control incidents.

Privilege	Subcategory of privileges	Includes privileges to	Actions that can be performed
Incidents (Genetec Mission Control™)	Add incident comments	Edit and delete incident comments	
	Add incident reminders	Modify and delete incident reminder	
	Change profile assignment		
	Edit incident filter templates		
	Export incidents		
	Forward incidents		
	Link incidents		
	Monitor incidents set to Testing mode		
	Take incident ownership	Attach documents	Delete attached documents.
		Add links	Delete attached links.
		Change additional fields	
		Change incident description	
		Change incident priority	
		Change incident type	
		Forcibly close unresolved incidents	
		Forcibly resolve incidents that have non-validated steps	
		Override incident ownership	

Privilege	Subcategory of privileges	Includes privileges to	Actions that can be performed
	Third-party integration mode		
	Transfer incidents		
	Transfer to a profile		
	Trigger incidents		
	TTR Metrics		

Task privileges

Task privileges control the accessibility of the tasks.

Privilege	Subcategory of privileges	Includes privileges to	Actions that can be performed
Administration	Incident configuration (Genetec Mission Control™)		Run the *Incident configuration* task in Config Tool.
Operation	Incident monitoring (Genetec Mission Control™)		Run the *Incident monitoring* task in Security Desk.
View public tasks	Modify columns		Modify the incident columns of a public task.
Investigation	Incident report (Genetec Mission Control™)	Modify reported incidents (Genetec Mission Control™)	
	Profile report (Genetec Mission Control™)		Review profile assignments.

Configuring access rights and privileges to workstations

Using the *Workstation management* task, you can provide access rights and privileges to selected users when they log on to Security Center through specific workstations.

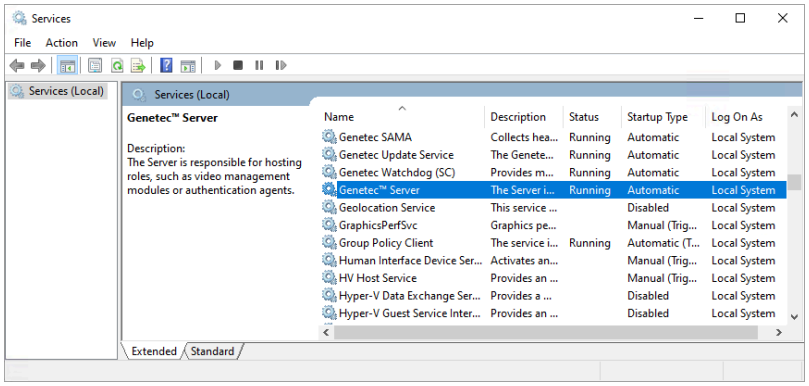
For more details on configuring the *Workstation management* task, see Granting extra access rights and privileges to specific workstation users.

Uninstalling Mission Control

You can uninstall Genetec Mission Control™ from the Windows Control Panel.

To uninstall Mission Control

1. Open the Windows Services application and stop the Genetec™ Server service.



2. Go to **Start** > **Control Panel** > **Add or Remove Programs**.
3. Select Mission Control Installer <*version*>.
   CAUTION: Mission Control Web API and Document Service are also uninstalled.

   If you want to uninstall only Mission Control, select Mission Control<*version*>
4. Click **Uninstall**.
5. Restart the Genetec Server service.

To uninstall the Mission Control patch release

To uninstall a component from Windows XP:

1. Click **Start** > **Control Panel** > **Add or Remove programs**.
2. Click **Show Updates**, select the patch release, and click **Remove**.

To uninstall a component from Windows 10:

1. Click **Start** > **Control Panel** > **Programs and Features**.
2. Click **View Installed Updates**, right-click the patch release, and select **Uninstall**.

To uninstall a component from Windows 8 or Windows Server 2012:

1. Right-click the **Start** button, and select **Programs and Features**.
2. Click **View Installed Updates**, right-click the patch release, and select **Uninstall**.

NOTE: Reboot your system to complete the uninstall.

To uninstall Mission Control in silent mode

1. Run the following command line from the Mission Control installation package:

   ```
   "MC Packages\MCInstaller.exe" /silent /remove
   ```

   CAUTION: This uninstalls Mission Control Web API as well.

   To uninstall only Mission Control in silent mode, run the following command from the installation package:

   ```
   msiexec /x "MC Packages\MissionControl\Genetec Mission Control <version>.msi"
   ```

Mission Control Web API and UI SDK

The Genetec Mission Control™ Web API enhances collaborative decision management by integrating custom functionality while the UI SDK is used to develop custom UI solutions.

For information regarding Mission Control Web API and UI SDK, visit Genetec Developer to find all resources regarding its configuration.

5

Staging to production environment

This section includes the following topics:

- "Phased approach to incident configuration deployment" on page 82
- "Considerations when importing incident configurations" on page 83

Phased approach to incident configuration deployment

You can deploy Genetec Mission Control™ incident configurations using a phased approach or in a Development, Testing, Acceptance and Production (DTAP) environment. This involves exporting incident type configurations between different Mission Control system environments.

Recommended deployment scenario

If you have a working Security Center environment and need to upgrade to a collaborative decision management system, the ideal deployment procedure is as follows:

1. Back up your Security Center production environment.
2. Create a new test environment.
3. Restore Security Center in the test environment.
4. Configure Mission Control in the test environment.
5. After testing is completed, export the Mission Control incident type configurations from the test environment into the production environment.

Considerations when exporting incident configurations

You can export and reuse Mission Control incident type configurations between multiple system environments:

- From development to testing environments
- From testing to acceptance environments
- From acceptance to production environments

Mission Control incidents can be configured to include Security Center events and entities such as alarms and more. While details of incident types are stored in the Mission Control database, details of Security Center entities are stored in the Directory database. The system assigns each entity its own Globally Unique Identifier (GUID), which is used to identify the entity associated to an incident type.

Consider a Mission Control incident called *Evacuate - Fire alert*, configured to be triggered when the Security Center alarm *Fire* is active. Both the incident and the alarm will have unique GUIDs assigned to them by the system.

If you delete the *Fire* alarm and create a new alarm with the same name and configuration settings, the GUID assigned to the new alarm by the system will still be different. The new alarm will no longer be associated with the *Evacuate - Fire alert* incident. You have to manually associate the new *Fire* alarm with the *Evacuate - Fire alert* incident.

This concept applies to incidents and entities in different system environments. Even though the names and configurations are same, the GUIDs will be different when exporting incidents between systems. You must therefore manually configure the entity and their incident associations after importing them from another system environment.

You can circumvent this process by using the backup Directory database from the source Mission Control environment and use it to restore the database in the target environment. This will ensure that previous configurations in the target environment are replaced with the same configurations and GUIDs in the backup directory from the source environment.

Considerations when importing incident configurations

To ensure that imported incident configurations, include the same associations to entities as the source system. The GUID details of the entities in the target and source systems must be identical.

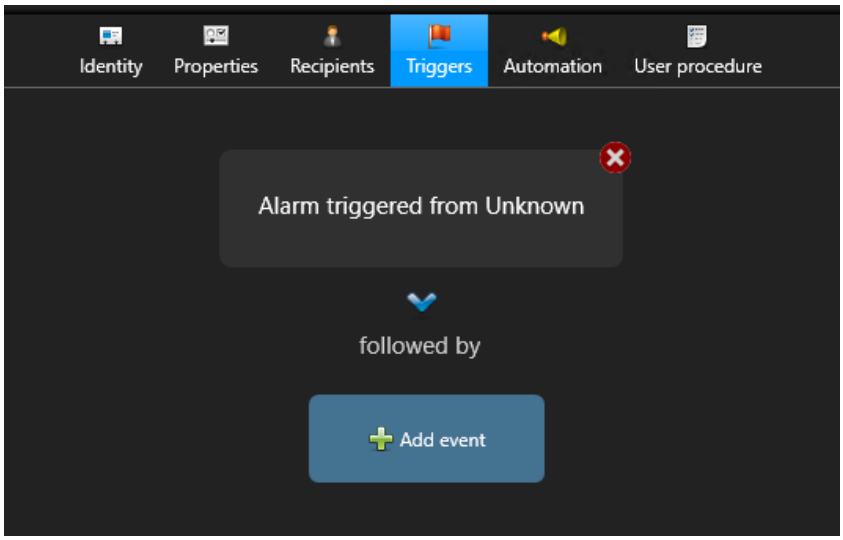
Importing incident configurations with attached entities

When you export incident configurations from one system to another, the JSON file includes all details of the Genetec Mission Control™ incidents and only the GUID information of the Security Center entities associated with the incidents.

When you import these incident configurations into your target system, the Mission Control import wizard looks for attached entities and other configurations by their GUID details as configured in the source system.

If entities with the same GUIDs exist in the target system, then the incident is imported and its entities are automatically associated in the target system.

However, if the entities in the target system have a different GUID, the incident type is imported by the import wizard but no entities are attached. It shows the entity as *Unknown* in the incident configuration.



Troubleshooting *Unknown* error when importing incident type configurations

Depending on your incident type configuration and where the entities are used, an *Unknown* error might affect the incident automation workflow, triggers, or recipients configuration in the target system.

When this happens, you must manually configure entities for incidents in the target system.

Typically, when importing incident type configurations, the following configurations might generate an *Unknown* error in the target system:

1. Alarms and other Security Center entities (such as cameras, doors, and so on)
2. Macros
3. Incident states
4. Incident priorities
5. Users, user groups, and associated privileges
6. Documents
7. Profiles
8. Federated entities (when federated from different sites)

9. Schedules
10. Threat levels

To ensure that there is no disconnect, it is recommended to verify the associated entity configurations in the target system in the listed order.

Exceptions when importing entity configurations

There are a few exceptions when importing configurations:

- **Profiles:** Profile details are stored in the Incident Manager database. They cannot be exported or imported between different Mission Control systems. You must assign incidents to profiles in the target system manually as this association cannot be copied between systems.
- **Additional fields:** You must ensure that the names of the additional fields in the target and source systems are identical.
- **Federated entities:** GUIDs for federated entities are obtained from the federated site. If the source and target Mission Control systems are federating the same system, the GUID of the entities in both systems will be identical.
- **Users and user groups:** You must ensure that the GUID of the user or user groups in the target and source system are identical. If the required user or user group does not exist, you must create one to configure details such as recipients for incident dispatch.
- **Incident states and priority:** Default system incident states and priorities have the same GUIDs irrespective of the system environment. You must ensure same GUIDs in source and target systems only if you are using custom states or priorities.
- **Documents:** If you have not used a shared network drive for documents related to incidents in both the source and target systems, you must copy the documents folder from the source to target system and manually configure documents for each incident type.

6

Upgrading Mission Control and RabbitMQ

This section includes the following topics:

- "Mission Control upgrading overview" on page 86

Mission Control upgrading overview

Upgrade your Genetec Mission Control™ environment based on your deployment architecture.

The following table lists the steps required to upgrade Mission Control.

<table>
<tr><th>Step</th><th colspan="2">Description</th></tr>
<tr><td colspan="3">Prerequisites</td></tr>
<tr><td colspan="3">Understand the supported upgrade path based on your Mission Control version and Document Service usage.</td></tr>
<tr><td colspan="3">Upgrade Mission Control</td></tr>
<tr><td>Download the latest build of Mission Control</td><td colspan="2">1. Visit the Product Download page in the Genetec Portal.
NOTE: You need a username and password to log on to the Genetec Portal.
2. From the Download Finder list, select your version of Security Center.
3. From the Mission Control section, download the required Mission Control package.</td></tr>
<tr><td rowspan="2">Upgrade Mission Control
NOTE: Genetec™ Incident Document Service, Genetec Mission Control™ Web API and RabbitMQ can be upgraded using the same installer.</td><td colspan="2">Validate the existing RabbitMQ configuration setup.</td></tr>
<tr><td colspan="2">Upgrade the main and failover servers.
NOTE: After upgrading to Mission Control 3.4.0.0 or later, a new Incident Document Service role with database DocumentService is created on all expansions servers if document service was enabled in your previous version of Mission Control.</td></tr>
<tr><td rowspan="4">Upgrade RabbitMQ
NOTE: If you are upgrading from the Genetec™ Update Service (GUS), RabbitMQ is not upgraded as a part of Mission Control and must be upgraded separately to the version available in the latest installer package.</td><td colspan="2">Validate the RabbitMQ configuration after upgrading the Mission Control server.</td></tr>
<tr><td colspan="2">RabbitMQ deployment architecture</td></tr>
<tr><td>Standalone machine</td><td>Server</td></tr>
<tr><td>Follow the below steps:
1. Deactivate the Incident Manager role.
2. Upgrade RabbitMQ.
3. Upgrade RabbitMQ nodes.
4. Restart the Genetec™ Server service on the Directory server.</td><td>Upgrade RabbitMQ on the server.</td></tr>
</table>

Supported upgrade paths to Mission Control 3.4.1.0

Mission Control requires a two-step upgrade to the latest software version if you are using Document Service.

Direct upgrade

Upgrade directly from these Mission Control releases only if you do not use Document Service. Confirm Document Service is unused before proceeding

- Mission Control 2.13.4.x (End of life)
- Mission Control 3.0.6.x
- Mission Control 3.1.3.x
- Mission Control 3.2.0.x
- Mission Control 3.2.1.x
- Mission Control 3.3.0.x
- Mission Control 3.3.1.x
- Mission Control 3.3.2.x
- Mission Control 3.3.3.x

CAUTION: If you use Document Service, upgrading from Mission Control 3.2.0.x or earlier does not migrate documents from MongoDB to the SQL database.

Two-step upgrade

If you use Document Service and are running Mission Control versions 3.2.0.x or earlier, a two-step upgrade is required to migrate all existing documents without any data loss. Ensure you meet this prerequisite before starting the upgrade process.

Performing this two-step upgrade is mandatory as its primary purpose is to ensure the system runs on the specific intermediate version long enough to complete the required data migration.

- **Step 1:** Upgrade to any of the Mission Control versions below:
  - Mission Control 3.2.1.x (Document Service 1.6)
  - Mission Control 3.3.0.x (Document Service 1.7)
  - Mission Control 3.3.1.x (Document Service 1.8)
  - Mission Control 3.3.2.x (Document Service 1.8)
  - Mission Control 3.3.3.x (Document Service 1.8)

  The Incident Manager role must be configured and online on this intermediate version to trigger the data migration process as the installer does not perform this migration.

  Allow the role to process and migrate the existing documents from MongoDB to the SQL database before proceeding to the next step.
- **Step 2:** Upgrade to Mission Control 3.4.1.0 (Incident Document Service 25.10.0.2).

Older version upgrades (Mission Control releases before version 2.13.4.x)

For information about upgrading from older versions of Mission Control, see the supported upgrade paths from earlier versions of Mission Control.

Upgrading Mission Control on the server

Follow the procedure below to upgrade Genetec Mission Control™ on the server.

Before you begin

- Understand the supported upgrade paths based on your Mission Control version and Document Service usage.
- Get the latest build of Mission Control. For information on how to get the package, see the Mission Control installation overview on page 54.

What you should know

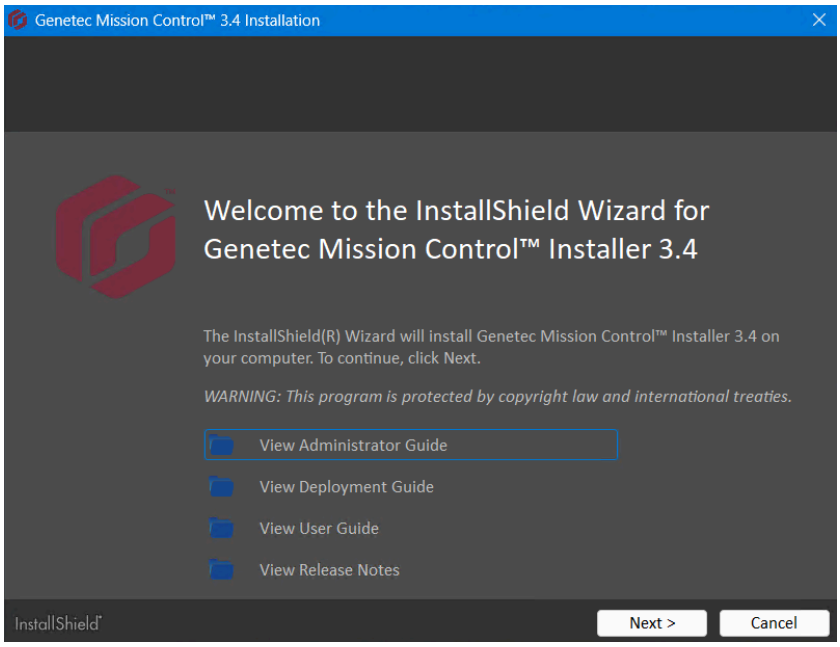
This process upgrades Mission Control, Genetec™ Incident Document Service, and RabbitMQ if it is on the same server as Mission Control. If you have Genetec Mission Control™ Web API installed, it is automatically upgraded to a compatible version when upgrading Mission Control.

Procedure

1 Launch the Mission Control Installer.

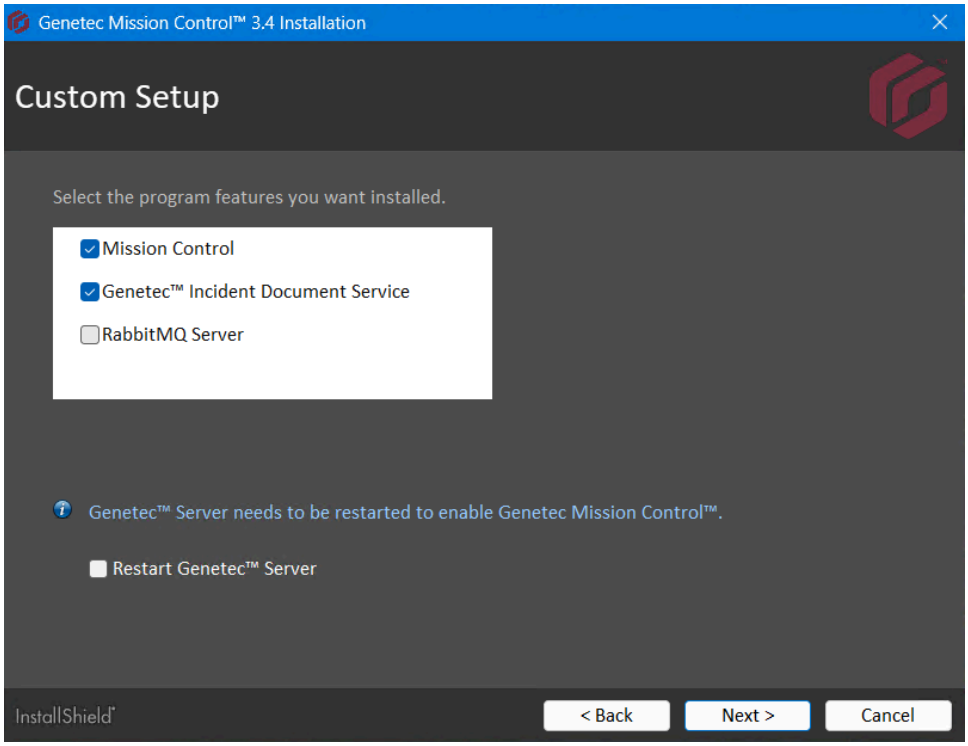
2 On the *Choose Setup Language* page, select English or French and click **Next**.

3 On the *Welcome to the InstallShield Wizard* page, click the links to read Mission Control documentation, and then click **Next**.



4 On the *License Agreement* page, read the terms in the *Genetec™ Software License Agreement*, select **I accept the terms in the license agreement**, and then click **Next**.

5 In the *Custom Setup* page, clear the **RabbitMQ Server** and **Restart Genetec™ Server** options and click **Next**.



NOTE: If you have installed RabbitMQ on this server you can select it in the dialog box to update it along with Mission Control.

6 Review the Mission Control Web API Configuration and Genetec Incident Document Service configuration before upgrading and do not restart the Genetec™ Server manually.

NOTE: After upgrading to Mission Control 3.4.0.0 or later, a new Incident Document Service role with database **DocumentService** is created on all expansions servers if document service was enabled in your previous version of Mission Control.

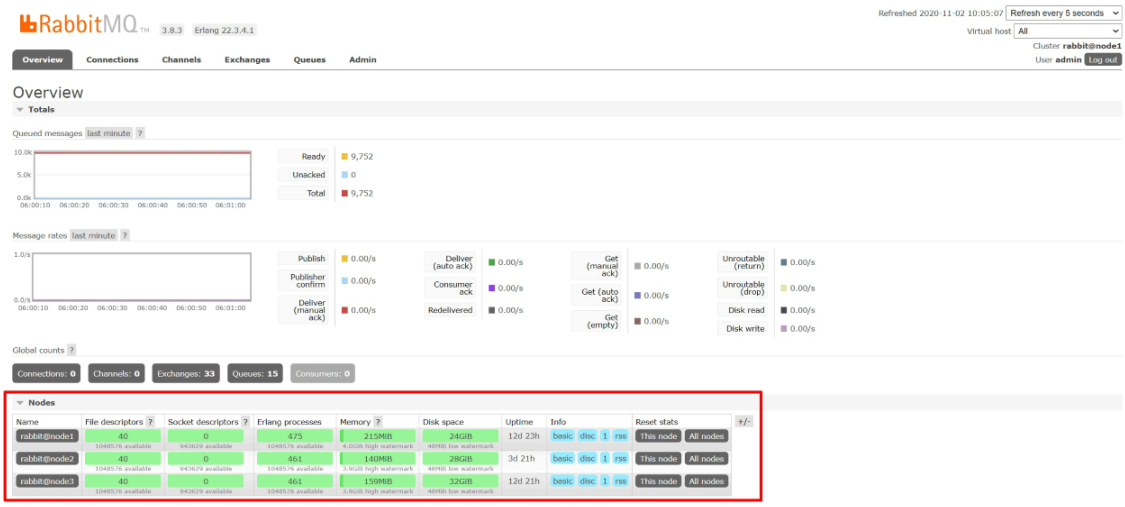
Validating RabbitMQ upgrade

If you are upgrading your Genetec Mission Control™ environment, it is recommended to validate your RabbitMQ setup before and after the upgrade.

Procedure

1 Open the RabbitMQ web portal (https://localhost:15671).

2 In the **Overview** tab, ensure that all RabbitMQ nodes are green. This indicates that the nodes are active within the cluster.



Your RabbitMQ configuration has been validated.

Upgrading RabbitMQ

If RabbitMQ is deployed on a standalone machine or in a high availability configuration with multiple nodes, you must use an adapted procedure to upgrade RabbitMQ.

Before you begin

- Validate your existing RabbitMQ configuration.
- Ensure that you are deploying RabbitMQ by the Genetec Mission Control™ installer. For external deployment of RabbitMQ with third-party solutions, refer the Upgrading RabbitMQ article.
- Copy the Mission Control 3.4.1.0 installation package to all RabbitMQ servers.
- Refer the RabbitMQ Clustering Guide to see how *.erlang* cookie paths are constructed.

TIP: It is recommended to use a 3-node cluster for high availability. For details, see Configuring high availability for RabbitMQ on page 39.

Procedure

1 If you are upgrading from RabbitMQ version 3.11.15 or later, enable feature flags by any of the below methods:

- RabbitMQ web portal
  a. Open the RabbitMQ web portal (https://localhost:15671)
  b. In the *Admin* tab click **Feature Flags**.
  c. Enable all feature flags and close the RabbitMQ web portal.
- RabbitMQ Command Prompt
  a. Open the RabbitMQ Command Prompt(sbin dir) and run this command:

```
rabbitmqctl.bat enable_feature_flag all
```

2 If you are upgrading from RabbitMQ 3.9.x or an earlier version , you need to install RabbitMQ 3.11.5 first by downloading the standalone installer from the Product Download page of the Genetec Portal.

NOTE: Enable all feature flags for RabbitMQ 3.11.5 before upgrading to the required RabbitMQ version.

3 Stop the RabbitMQ service.

4 Back up the *.erlang.cookie* file to a temporary storage location.
Usual file path for the *.erlang.cookie* is *C:\Windows\ServiceProfiles\LocalService*.

5 Paste the *.erlang.cookie* file from the temporary storage location to the following folders:

- *C:*
- *C:\Users\<username>*
- *C:\Windows\System32\config\systemprofile*

NOTE: If an *.erlang.cookie* file exists in the destination folder, overwrite it with the copied file.

6 If using your own certificates in *C:\Program Files (x86)\Genetec\RabbitMQ*, you must create a backup for them as well.

7 Use the *Custom setup* in the Mission Control 3.4.1.0 installation wizard to upgrade RabbitMQ.

8 Restart the RabbitMQ service.

Upgrading RabbitMQ on the server

If your deployment architecture includes RabbitMQ on a dedicated server, use the following procedure to upgrade your system.

Procedure

1 Enable feature flags by any of the below methods:

- RabbitMQ web portal
  a. Open the RabbitMQ web portal (https://localhost:15671)
  b. In the *Admin* tab click on **Feature Flags**.
  c. Enable all feature flags and close the RabbitMQ web portal.
- RabbitMQ Command Prompt
  a. Open the RabbitMQ Command Prompt(sbin dir) and run this command:

```
rabbitmqctl.bat enable_feature_flag all
```

2 If you are upgrading from RabbitMQ 3.9.x or an earlier version, run the RabbitMQ 3.11.5 standalone installer from the Genetec Portal Product Download page and enable all its feature flags by repeating Step 1. If you are upgrading from RabbitMQ version 3.11.15 continue with the below steps.

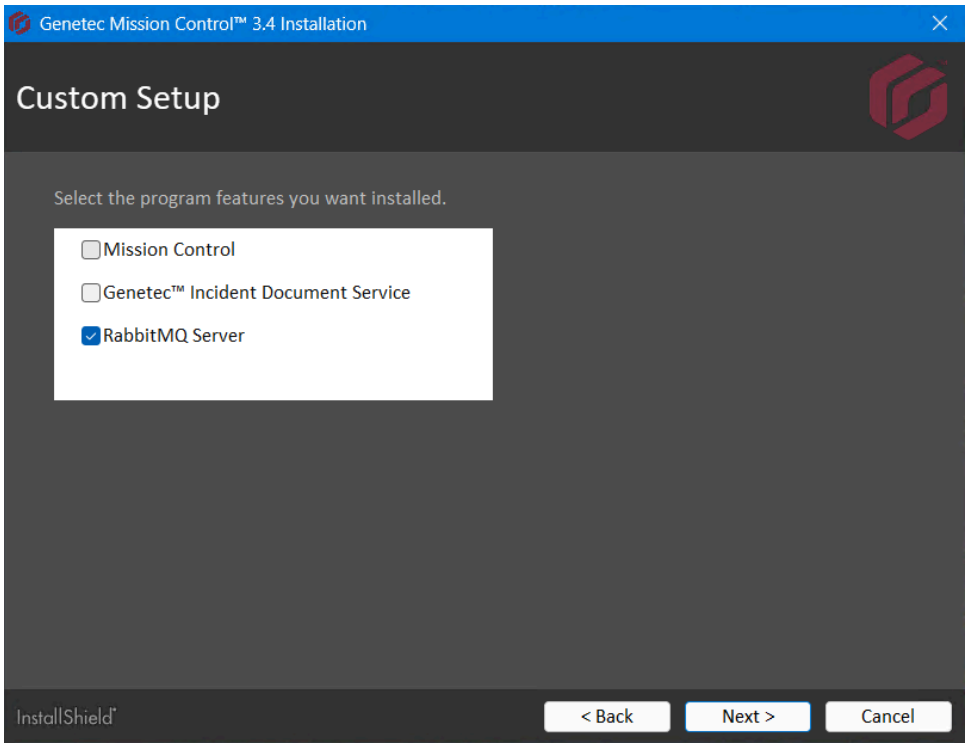
3 Launch the Mission Control 3.4.1.0 Installer.

4 On the *Choose Setup Language* page, select English or French and click **Next**.

5 On the *Welcome to the InstallShield Wizard* page, click the links to read Mission Control documentation, and then click **Next**.

6 On the *License Agreement* page, read the terms in the *Genetec™ Software License Agreement*, select **I accept the terms in the license agreement**, and then click **Next**.

7 On the *Custom Setup* page:
- Select the **RabbitMQ Server** option.
- Clear both the **Mission Control** and **Genetec™ Incident Document Service** options.



8 Click **Install** to complete the RabbitMQ upgrade.

Security Desk and the Mission Control *Incident monitoring* task remain online during the RabbitMQ upgrade. The RabbitMQ cluster configuration is not disturbed during this upgrade.

Upgrading RabbitMQ nodes

Upgrading the nodes of your RabbitMQ cluster can be done without any downtime for your system.

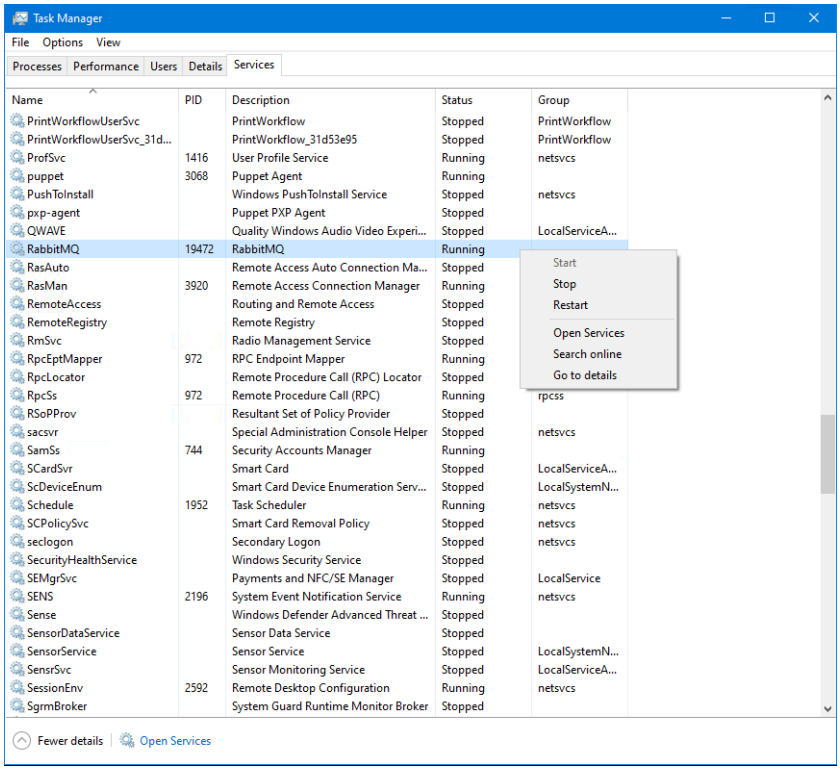
Before you begin

- Validate your RabbitMQ upgrade to ensure optimal efficiency.
- Refer the Upgrading RabbitMQ article to understand the deployment process.
- Make sure the Incident Manager role and the RabbitMQ cluster is configured correctly.
- Copy the Genetec Mission Control™ installation package to all RabbitMQ servers.

Procedure

1 Create a copy of the *RabbitMQ.conf* file and the certificates folder for the RabbitMQ server on every system. The folder location is *C:\Program Files (x86)\Genetec\RabbitMQ*.

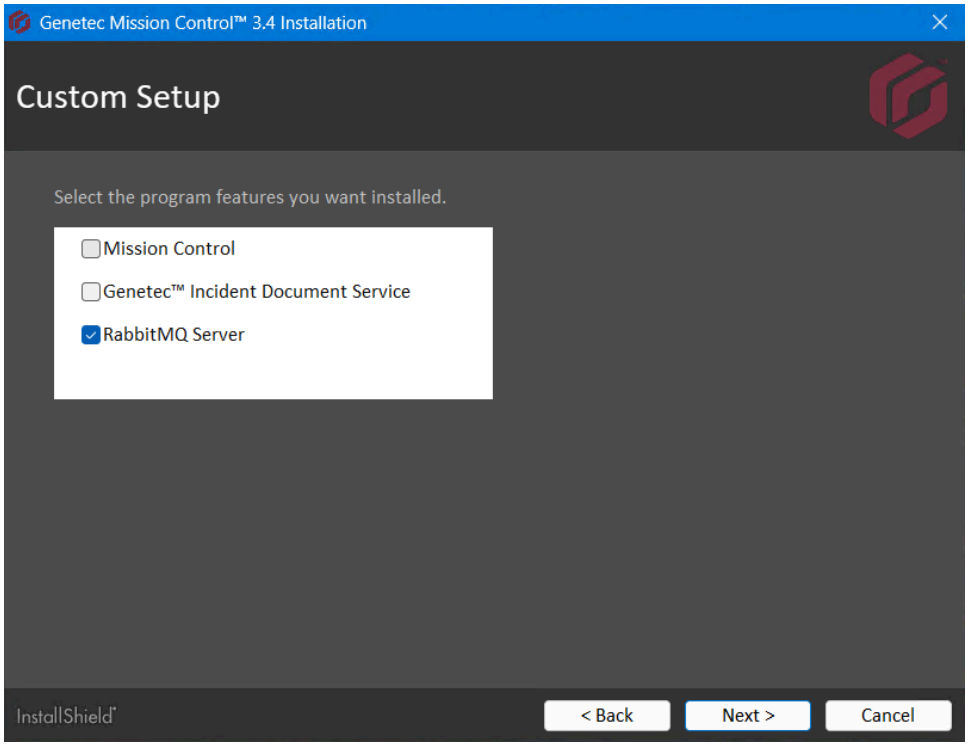
2 Open the *Task Manager*, in the *Services* tab right-click on **RabbitMQ** and stop the service.



3 Run the *Genetec Mission Control™ Installer*.

4 On the *Custom Setup* page:

- Select the **RabbitMQ Server** option.
- Clear both the **Mission Control** and **Genetec™ Incident Document Service** options.



5 Click **Install** to complete the RabbitMQ upgrade.

NOTE: During the install, it's expected that the Incident Manager role and the RabbitMQ manager web page display warning messages.

6 After the installer finishes its process, open *Task manager* and stop the RabbitMQ service.

7 In the file explorer, delete the original certificate folder and rename the copied folder with the name of the original.

8 Delete the original *RabbitMQ.conf* file and restore the copied file in its place.

9 Start the RabbitMQ service in the *Task Manager*.

The Incident Manager role and the RabbitMQ manager web page do not display warning messages.

10 Repeat the above process on all remaining RabbitMQ servers.

Restarting services

You can restart Genetec™ Server and RabbitMQ service from the *Microsoft Management Console Services*.

Procedure

Restarting Genetec™ Server service:

1 On the server, open *Microsoft Management Console Services* (`services.msc`).

2 Select the Genetec™ Server from the list of services and click **Restart**.

Security Desk remains offline until Genetec™ Server is back online.

Restarting RabbitMQ service:

1 On the RabbitMQ server, open *Microsoft Management Console Services* (`services.msc`).

2 Select the RabbitMQ from the list of services and click **Restart**.

Part II

Mission Control mobile access

This part includes the following chapter:

- Chapter 7, "Using Mission Control on Genetec Mobile" on page 96

7

Using Mission Control on Genetec Mobile

This section includes the following topics:

- "About Mission Control on Genetec Mobile " on page 97

About Mission Control on Genetec Mobile

You can perform many of the Security Desk Incident monitoring and investigation functions from your mobile device using Genetec™ Mobile.

To use Genetec Mission Control™ features on Genetec Mobile, you must do the following:

- Ensure you've configured the Mobile Server role as specified in the Security Center Administrator Guide.
- Ensure that you have a compatible version of the Genetec Mission Control Web API installed on the server that hosts the Incident Manager role.

  NOTE: To access Mission Control features on the mobile app, the minimum requirement is Mission Control 2.13.4.0 and Mission Control Web API 2.0.
- Install Genetec Mobile on your mobile devices (Android and iOS).
- Configure your networks (Wi-Fi, VPN, and so on) so that the mobile devices can connect to the Mobile Server role.
- Ensure that your license supports mobile devices (**Number of mobile devices** > 0). To use Mission Control on Genetec Mobile, you must have a valid Mission Control Operator license.

  NOTE: Each time a user with Mission Control privileges logs in on a mobile device, it uses one operator license. However, if they are already logged in on Security Desk or the Genetec™ Web App, an additional license is not required.
- Enable Mission Control features in the mobile app:

  1. Log on to the mobile app.

     Administrator credentials are required to log on for the first time.
  2. In Config Tool, enable the mobile feature flag:

     a. Go to **System** > **Roles** > **Mobile Server** > **Properties**.

     b. In the *Features* section, select the **Mission Control** option.

        If the **Mission Control** option doesn't exist:

        1. Click  and add the **Mission Control** option.
        2. In the *Feature key* dialog box that opens, enter `Mission Control`.
        3. Click **Add**.

Related Topics

Security Center streaming capacities for Genetec Mobile

Product compatibility for Mission Control

Glossary

allowed user

An allowed user of a Genetec Mission Control™ profile is a user to whom the profile can be assigned, as long as the user is connected to Security Center through Security Desk.

automation

Automation is a series of activities associated to an incident type. These activities are performed by the system during the life cycle of an incident. The activities can change the incident state and properties, affect other entities in the system, or simply wait for a condition to become true. The automation workflow helps automate simple tasks, such as exporting the incident when it is resolved, so operators can focus on more complex ones. (Replaces the term incident workflow)

collaborative incident

A collaborative incident is an incident type that requires the collaboration of multiple teams to resolve. Each team has specific tasks to follow, which are represented by sub-incidents. The collaborative incident is resolved when all its sub-incidents are resolved.

dynamic user procedure

A dynamic user procedure (or dynamic SOP) is a standard operating procedure where the step sequence depends on the response provided by the operator at each step. The dynamic SOP is represented by a flowchart in Config Tool.

failover

Failover is a backup operational mode in which a role (system function) is automatically transferred from its primary server to a secondary server that is on standby. This transfer between servers occurs only if the primary server becomes unavailable, either through failure or through scheduled downtime.

Federation™

Federation™ joins multiple, independent Genetec™ security systems into a single virtual system. With this feature, users on a central system, called the Federation host, can view and control entities that belong to remote systems.

Genetec Mission Control™

Genetec Mission Control™ is a collaborative decision management system that provides organizations with new levels of situational intelligence, visualization, and complete incident management capabilities. It allows security personnel to make the right decision when faced with routine tasks or unanticipated situations by ensuring a timely flow of information. To learn more about Genetec Mission Control™, refer to the Genetec™ resource center.

Genetec Mission Control™ Web API

The Genetec Mission Control™ Web API is a suite of development tools with which our partners and customers can enhance their collaborative decision management systems by integrating additional software and custom functionality.

high availability

High availability is a design approach that enables a system to perform at a higher than normal operational level. This often involves failover and load balancing.

incident (Genetec Mission Control™)

A Genetec Mission Control™ incident is an undesirable or unusual situation that needs investigation and resolution, or a routine, scheduled task that requires monitoring.

incident category

An incident category is an entity that represents a grouping of incident types that have similar characteristics.

Incident configuration

The *Incident configuration* task is an administration task that you can use to configure the incident types, the incident categories, and the support documents for Genetec Mission Control™. You can also use this task to generate reports on the changes made to incident types.

Incident Document Service

The Incident Document Service role is responsible for storing and managing supporting documents and forms for incident response. The role uses a configurable API port to upload and download documents.

Incident Manager

The Incident Manager is the central role that recognizes situational patterns, and triggers incidents in a Genetec Mission Control™ system. This role manages the automation workflows and keeps track of all user activities that are related to incidents.

Incident monitoring

The *Incident monitoring* task is an operation task that you can use to monitor and respond to incidents. From this task, you can see the incidents displayed on a map, thus improving your situational awareness.

incident owner

The incident owner is the incident recipient who took ownership of the incident. Only the incident owner can take actions to resolve the incident. An incident can only have one owner at a time.

incident recipient

An incident recipient is a user or user group that the incident has been dispatched to. Incident recipients can see the incident in the *Incident monitoring* task.

Incident report

The *Incident report* task is an investigation task that you can use to search, review, and analyze Genetec Mission Control™ incidents.

incident supervisor

An incident supervisor is a user who sees an incident in the *Incident monitoring* task because they supervise the incident recipients. Incident supervisors are not incident recipients themselves. A user cannot be both supervisor and recipient of the same incident.

incident trigger

An incident trigger is an event or a sequence of events that can trigger an incident. The Genetec Mission Control™ Rules Engine looks for specific combinations of events (type, time, correlation, and frequency) to determine whether to trigger an incident.

incident type

An incident type entity represents a situation that requires specific actions to resolve it. The incident type entity can also be used to automate the incident detection in Genetec Mission Control™ and to enforce the standard operating procedures that your security team must follow.

Monitoring

The *Monitoring* task is an operation task that you can use to monitor and respond to real-time events that relate to selected entities. Using the *Monitoring* task, you can also monitor and respond to alarms.

profile

A profile in Genetec Mission Control™ is an incident filter used to distribute active incidents among the operators who are currently logged on to the system. Profiles filter incidents by type and by occurrence location.

Rules Engine

The Rules Engine is the component of the Genetec Mission Control™ system that analyzes and correlates the events collected by Security Center, based on predefined rules. The Rules Engine uses these events to detect and trigger incidents in the Genetec Mission Control™ system.

simple user procedure

A simple user procedure (or simple SOP) is an ordered list of steps that do not involve multiple-choice questions. Operators might be allowed to resolve the incident without validating all the steps, or to complete the steps out of sequence, depending on the configuration.

standard operating procedure

A standard operating procedure (SOP) or user procedure is a series of steps that the operator must follow before changing an incident state to *Resolved*. Each step can range from a simple task, such as *Call 911*, to a complex task that requires support documents to accomplish. All actions taken by the operator are logged by the system in the incident history for audit purposes.

threat level

A threat level is a predefined scenario that you activate for a specific area or the entire system to manage special situations, such as a business event or emergency evacuation. Activating a threat level triggers a series of actions tailored to the scenario. Deactivating the threat level returns the system to regular operation.

Time To Response or Time To Resolution

Time To Response or Time To Resolution (collectively TTR) are metrics used in Genetec Mission Control™ that refer to how long it takes to respond to incidents, and resolve incidents and dynamic standard operating procedure steps.

user procedure

Replaces the term "incident procedure". See "standard operating procedure".

validated step

A validated step in an standard operating procedure (SOP) is a step that is either completed or skipped with a justification.

Vault

The Vault is a tool that displays your saved snapshots and exported G64, G64x, and GEK (encrypted) video files. From the Vault, you can view the video files, encrypt and decrypt files, convert files to ASF, or package files with the Genetec™ Video Player.

Where to find product information

You can find our product documentation in the following locations:

- **Genetec™ TechDoc Hub:** The latest documentation is available on the TechDoc Hub.

  Can't find what you are looking for? Contact documentation@genetec.com.
- **Installation package:** The Installation Guide and Release Notes are available in the Documentation folder of the installation package. These documents also have a direct download link to the latest version of the document.
- **Help:** Security Center client and web-based applications include help, which explains how the product works and provide instructions on how to use the product features. To access the help, click **Help**, press F1, or tap the **?** (question mark) in the different client applications.

Technical support

The Genetec™ Technical Assistance Center (GTAC) provides technical support to customers worldwide. As a Genetec customer, you have access to TechDoc Hub, where you can find documentation and answers to product questions.

- **Genetec TechDoc Hub:** TechDoc Hub includes articles, manuals, and videos that help answer questions and resolve technical issues. Before contacting GTAC or opening a support case, search TechDoc Hub for potential fixes, workarounds, or known issues.

  To access TechDoc Hub, sign in to the Genetec Portal and click Go to TechDoc Hub. If you can't find what you're looking for, contact documentation@genetec.com.
- **Genetec Technical Assistance Center (GTAC):** Information about contacting GTAC is available in the Genetec Advantage Description.

Technical training

In a professional classroom environment or from your own office, qualified trainers guide you through system design, installation, operation, and troubleshooting. Technical training is available for all products and for customers with varying levels of technical experience, and it can be customized to meet specific needs and objectives. For more information, see the Genetec training certification courses directory.

Licensing

- For license activation or reset requests, contact GTAC at https://www.genetec.com/portal/support.
- For issues with license content, part numbers, or orders, contact Genetec Customer Service at customerservice@genetec.com or call 1-866-684-8006 (option 3).
- For demo licenses or pricing questions, contact Genetec Sales at sales@genetec.com or call 1-866-684-8006 (option #2).

Hardware product issues and defects

Contact the Genetec Technical Assistance Center (GTAC) at https://www.genetec.com/portal/support to report or resolve issues with Genetec appliances or hardware purchased through Genetec Inc.